# Harder Benchmarks Are Not Always Better: Evaluating LLM Distractors for Multiple-Choice Benchmarks

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## Abstract

In multiple-choice question answering (MCQA), model abilities are outpacing researchers' ability to make new benchmarks that test them. Researchers, in turn, accelerate MCQA benchmark design by using LLMs to create wrong answer options, called distractors. While work claims this makes benchmarks harder, we challenge the idea that difficulty alone captures distractors' usefulness for NLP evaluation. We test this by using three LLMs over three MCQA benchmarks for two tasks: *distractor generation* from question–answer pairs and *distractor extension* from existing MCQs. We score MCQs via metrics in education (difficulty, discriminability, and writing quality), exposing that: 1) LLM distractors often make items harder than human-guided ones, but these items are rarely more discriminative and can sacrifice quality; 2) adding distractors to an MCQ can lower model accuracy, but fails to better separate model ranks; and 3) LLMs struggle to balance plausibility, option length, and simulated mistakes—areas for improvement that humans can monitor. Overall, we argue that model accuracy alone can miss what makes MCQA distractors useful. From our analyses, we outline best practices for how to use LLMs in MCQA benchmark construction.[1]

## 1 Introduction: The Art of Distraction

Several NLP benchmarks rely on multiple-choice questions (MCQs; Clark et al., 2020): models select the best answer to a question while avoiding the plausible but wrong answers, called distractors. MCQA is attractive as it is easy to score, adaptable to many domains, and mirrors real student exams (Balepur et al., 2025). LLMs are rapidly improving at MCQA (Singh et al., 2025), so researchers must quickly create new MCQA benchmarks if they want to challenge current LLMs (Akhtar et al., 2026).

[1]We release our code and data at: https://anonymous.4open.science/r/mcqa-distractor-supplement-2317




Figure 1: Two failures when LLMs generate distractors on ARC's multiple-choice benchmark: item (a) is hard for the wrong reason, as 10/14 test-takers pick a distractor that is right; item (b) is too easy as no test-taker picks GPT's long, obvious distractors. Numbers in parentheses are how many models pick that choice.

A bottleneck for quickly creating MCQA benchmarks is its distractors—one of the hardest parts of MCQ writing (Lai et al., 2016). Distractors that are too similar to the gold answer can unfairly penalize test-takers (Figure 1, a), while ones clearly wrong make MCQs too easy (Figure 1, b). Educators adhere to rigorous writing guidelines when authoring distractors to ensure difficulty and quality (Haladyna et al., 2002), but this human effort is infeasible to scale to the thousands of questions MCQA benchmarks demand (Hendrycks et al., 2021).

To alleviate these burdens, NLP researchers have started prompting LLMs to generate distractors for MCQA benchmarks via two main approaches. First, some produce distractors given an MCQ's question and gold answer, aiming to match the difficulty and

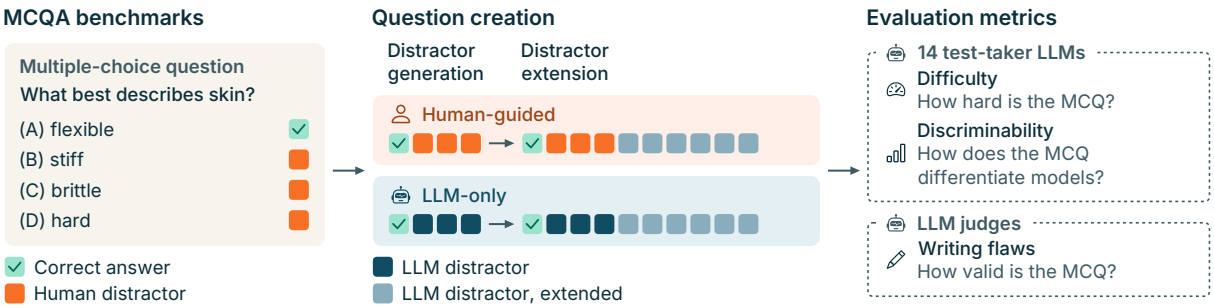



Figure 2: **Evaluation Framework Overview.** We compare MCQA distractors generated with human guidance versus LLMs alone in two tasks: 1) *distractor generation* from a question–answer pair, where human-guided items are the original MCQA benchmark items; and 2) *distractor extension* from an existing MCQ, adding six new distractors to either the human-written or LLM-written MCQ. We operationalize the usefulness of MCQs based on three metrics from education research: difficulty for 14 test-taker LLMs, discriminability of model ranks, and writing quality.

quality of human-authored distractors (Zhang and VanLehn, 2021; Akter et al., 2026). Second, others add extra distractors to existing MCQs to scale difficulty while preserving quality. MMLU-Pro is an example (Wang et al., 2024); it adds six distractors to MMLU, spurring a 16–33% LLM accuracy drop.

While prompting LLMs to write distractors for MCQA benchmarks over human authoring is common, work often claims LLM distractors are useful because they lower model accuracy (Wang et al., 2024). However, lower accuracy does not mean a benchmark is better: it could reflect unanswerable questions (Figure 1) or fail to distinguish between models, leaving their relative ranks unchanged and offering little extra insight into LLM abilities. Thus, our paper aims to test the usefulness of LLM distractors for MCQA benchmarks beyond just accuracy.

Our evaluation starts by prompting three generator LLMs (GPT-5.2, Gemini-3.1 Pro, Qwen-3.5 397B) to write distractors for three MCQA datasets (ARC, MMLU, GPQA) across two tasks: (1) *distractor generation*, where models create distractors from an input question and answer; and (2) *distractor extension*, where models add new distractors to an MCQ (Figure 2, center). In both tasks, we compare distractors created with human guidance (human-guided) versus LLMs alone (LLM-only).

Drawing on education research (Haladyna et al., 2002), we evaluate the usefulness of each MCQ for benchmarks using three metrics: (1) the item's difficulty; (2) how well the item can discriminate model ranks; and (3) how often the item follows best practices in writing quality (Figure 2, right). We measure (1) and (2) by fitting an Item Response Theory model (IRT; Lord and Novick, 2008) on 14 smaller, test-taker models on our MCQs. For (3), we apply LLM judges equipped with a 19-rule rubric from education to detect writing flaws (Tarrant et al., 2006). While most works focus just on difficulty, we include discriminability to evaluate an MCQ's impact on model ranks, and quality to test whether difficulty merely stems from poor writing.

We expose that LLM distractors often make items harder (7/9 cases in distractor generation), but these harder items are rarely more discriminative (2/7 cases) and can sacrifice writing quality (3/7 cases) (§4.1). Similarly, extra distractors can lower model accuracy but fail to consistently separate model ranks (§4.2), partially explained by test-takers just favoring early options (§4.3). LLM distractors also have writing flaws: relative to humans, LLMs have fewer implausible distractors, but GPT generates distractors that are plausibly right or obviously wrong due to distinct length (§5.2). We then study models' reasoning traces to understand how LLMs construct distractors: LLMs try to simulate mistakes and recall related facts—similar to educators—but often make shallow perturbations of the gold answer without aptly targeting test-taker errors (§5.3).

While LLM distractors seem easy to use, evaluating their usefulness is challenging: accuracy alone is insufficient. To guide future work, we distill our analyses into best practices for using LLM distractors in benchmark design (§6). We contribute:

1. An evaluation framework to analyze how well LLMs create distractors in MCQA benchmarks.
2. Evidence that LLM distractors can increase difficulty but rarely aid discriminability and can lower quality, that extra distractors lower accuracy without shifting LLM ranks, and that LLMs struggle to simulate expert mistakes.
3. A released dataset of 35,460 MCQs produced by three frontier LLMs and 496,440 responses from 14 smaller LLMs to aid future research.

4. A synthesized list of best practices for using LLMs during MCQA benchmark construction.

## 2 Related Work

To ground our evaluation framework, we synthesize literature across distractor generation (§2.1), synthetic data (§2.2), and education metrics (§2.3).

### 2.1 Distractor Generation

Distractor generation is a long-standing task, where researchers design models to help educators write MCQA exams (Alhazmi et al., 2024; Feng et al., 2024; Lee et al., 2025). Earlier models relied on heuristics and trained neural networks (Mitkov et al., 2003; Liang et al., 2018), with more recent work instead exploiting the strengths of LLMs via prompting (Bitew et al., 2023; Mucciaccia et al., 2025; Qu et al., 2024). Feng et al. (2024) created new methods for math-based educational MCQ distractor generation, finding models generate surface-level valid distractors but struggle with anticipating student errors. Lee et al. (2025) use student choice behavior to improve distractor plausibility while producing questions with high discrimination.

MCQA's use in NLP has recently motivated applying such methods to benchmark items (Zellers et al., 2018; Liu et al., 2022; Wang et al., 2024), where distractors must challenge models versus students. We bridge these two fields, drawing on education to score the usefulness of LLM distractors for NLP evaluation (Haladyna et al., 2002). Closest to us, Byun and Choi (2025) train models on MMLU to form distractors, but do not prompt stronger LLMs.

### 2.2 Synthetic Datasets and Benchmarks

Since collecting human-written text is costly, LLM-generated (i.e., synthetic) data is now commonly used to train models (Long et al., 2024; Li et al., 2025a; Nadăș et al., 2025) and build new evaluations (Wang et al., 2024; Zhang et al., 2025). Work increasingly tests the benefits and risks of such data (Shumailov et al., 2024; Liu et al., 2024), and we similarly audit LLM-generated distractors—a key part of an MCQ that determines the item's difficulty.

Similarly, humans used to be the ones to write MCQA benchmarks. Earlier datasets used exams written by educators (Clark et al., 2018; Hendrycks et al., 2021), and recent, harder benchmarks crowdsource questions from PhD experts (Rein et al., 2024; Phan et al., 2025). Today, LLMs can help generate MCQA benchmarks. AutoBencher frames benchmark creation as an optimization problem, using LLMs to create diverse datasets based on qualities such as salience, novelty, and difficulty (Li et al., 2025b). MMLU-Pro curated MMLU by removing noisy questions and expanding the answer set from four to ten options (Wang et al., 2024). We contribute to this line of work by studying whether LLM-generated distractors that make items harder also preserve quality desired for MCQA evaluation.

### 2.3 Applying Education Metrics in NLP

Education research has long developed metrics for measuring the usefulness of MCQA exams, which can be directly applied to NLP benchmarks. Item Response Theory (IRT; Lord and Novick, 2008) is a psychometrics tool for testing how items shape test-taker scores, often used to study difficulty (how hard items are) and discriminability (how well they discern students' skill). Work uses IRT in NLP by treating models as examinees (Lalor et al., 2016), studying dataset difficulty (Rodriguez et al., 2021; Kipnis et al., 2024; Hofmann et al., 2025) and discriminability (Vania et al., 2021; Sung et al., 2025).

Beyond metrics on test-taker performance, education research establishes quality guidelines for MCQ authoring (Haladyna and Downing, 1989; Haladyna et al., 2002). Work shows that items violating them—common even in high-stakes exams (Tarrant et al., 2006)—decrease IRT difficulty and discriminability (Schmucker and Moore, 2025). More recently, work designs toolkits to detect these flaws automatically (Moore et al., 2022, 2023; Balepur et al., 2026). Drawing on this line of work, we extend IRT modeling and quality metrics to study the usefulness of LLM distractors.

## 3 Evaluation Framework for Distractors

A multiple-choice question (MCQ) has a question stem $q$ (e.g., "What best describes skin?") and $n$ choices $\mathcal{C}$ (e.g., "flexible", "stiff", "brittle", "hard"), with a single correct answer $a \in \mathcal{C}$ (e.g., "flexible"). The other choices are distractors $\mathcal{D} = \mathcal{C} \setminus \{a\}$: incorrect but plausible answer choices that challenge test-takers by targeting common misconceptions.

The distractors are a key aspect of an MCQ (Lai et al., 2016), but generating high-quality, difficult ones via manual human effort is onerous. Recent work instead relies on LLMs to author distractors (Wang et al., 2024), but such works rarely evaluate how these distractors impact LLM evaluation. We address this with an evaluation framework for dis-

tractor usefulness. We now present our distractor creation tasks (§3.1) and input data for them (§3.2), followed by our models (§3.3) and metrics (§3.4).

### 3.1 Distractor Generation and Extension

To study the benefits and drawbacks of LLM distractors in MCQA benchmarks, we define two tasks that NLP researchers often perform with LLMs to automatically write distractors (Figure 2, center):

- **Distractor Generation (D-GEN):** For a question–answer pair (e.g., $q$ ="What best describes skin?", $a$ ="flexible"), return $n-1$ distractors $\mathcal{D}$ to form an MCQ $\mathcal{Q} = (q, \mathcal{D} \cup \{a\}, a)$. In our main experiments, our LLMs performing D-GEN generate $n-1=3$ distractors, creating a four-choice MCQ.
- **Distractor Extension (D-EXT):** For an MCQ $\mathcal{Q} = (q, \mathcal{C}, a)$, create $m$ extra distractors $\mathcal{D}'$ to form a new MCQ $\mathcal{Q}' = (q, \mathcal{C} \cup \mathcal{D}', a)$, heightening MCQ difficulty (Wang et al., 2024; Du et al., 2025). In our D-EXT experiments, we use four-choice MCQs and prompt LLMs to add $m=6$ new distractors, forming a ten-choice MCQ. This follows the exact protocol of MMLU-Pro, the most popular benchmark created via D-EXT (Wang et al., 2024).

### 3.2 Datasets for Evaluating LLM Distractors

We collect inputs for studying D-GEN and D-EXT via MCQs in three common benchmarks of varied difficulty: 1) **ARC-Challenge** (Clark et al., 2018), grade-school science exam questions drawn from US standardized exams; 2) **MMLU** (Hendrycks et al., 2021), 57 college-level topics based on public practice materials (GRE, USMLE, AP, etc.); and 3) **GPQA** (Rein et al., 2024), graduate-level biology, physics, and chemistry items written by PhD-level domain experts. Each benchmark has MCQs with $n=4$ choices. We sample 1000 MCQs each from ARC and MMLU and use all 448 MCQs from GPQA (details in Appendix A.1).

In D-GEN, LLMs generate distractors using the questions and answers from these benchmarks as inputs, letting us compare their usefulness against the original human-written distractors. In D-EXT, we compare adding distractors to original benchmark MCQs versus MCQs from D-GEN. For both tasks, we refer to MCQs containing human-written distractors as **human-guided**, and those with only model-generated distractors as **LLM-only**. These benchmarks may exist in LLM training data (Magar and Schwartz, 2022), but our claims hold even when filtering likely contaminated ARC items (§3.4).

### 3.3 Generator and Test-Taker Model Design

Our evaluation framework relies on two models: (1) **generator models** that write MCQ distractors; and (2) **test-taker models** that solve MCQs from (1). We focus on studying generator model abilities, while test-taker model responses form our metrics for distractors' impact on evaluation (§3.4). We discuss the generator and test-taker models next.

**Generator Models** Researchers typically rely on stronger LLMs to generate distractors (Wang et al., 2024), so we examine three frontier closed and open-weight LLMs as generator models: GPT-5.2 (OpenAI, 2025), Gemini-3.1 Pro (Google DeepMind, 2026), and Qwen-3.5 397B (Qwen Team, 2026). We prompt models to write three or six distractors for D-GEN and D-EXT, respectively, and return a JSON object with the distractors (Prompts A.3, A.4), based on prompts from Wang et al. (2024). We run generation in InspectAI (UK AISI, 2024), a standardized framework for LLM evaluations. We use a maximum generation length of 10,000 tokens and two retries on API errors. Appendix A.3 has full implementation details.

**Test-Taker Models** We select 14 LLMs to solve MCQs as test-taker models. For diversity, we use a closed-source and 13 open-weight models of varied sizes: GPT-5.4-mini,[2] Nemotron-2 (Basant et al., 2025, 9B), Gemma-3 (Team et al., 2025, 4B, 12B), Qwen-3 (Yang et al., 2025, 4B), Olmo-3 (Olmo et al., 2025, 7B, 32B), LLaMA-3 (Grattafiori et al., 2024, 3B, 8B), LFM-2 (Amini et al., 2025, 1.2B, 2.6B), EXAONE-4 (Bae et al., 2025, 1.2B), and Granite-4 (IBM Research, 2025, 1B, Micro).

Each test-taker gets the question stem and answer choices and returns a valid JSON with a single key, `answer`, whose value is one of the displayed choice letters (Prompt A.1)—adapting InspectAI's MCQA prompt (UK AISI, 2024). We shuffle MCQ choices to correct for positional bias (Wei et al., 2024; Appendix A.4). 98.84% of the 496,440 responses contain a valid choice letter, and if no valid answer letter is parsed (1.16%), we assign a random answer letter (Appendix A.2). We prompt each model one time, following frameworks like lm-evaluation-harness (Biderman et al., 2024), HELM (Liang et al., 2023), and OLMES (Gu et al., 2025). Appendix A.3 has full implementation details.

[2] https://tinyurl.com/gpt54-mini-api

### 3.4 Metrics for Multiple-Choice Usefulness

We aim to measure the *usefulness* of MCQs for NLP evaluation when distractors are human-guided or use LLMs alone. To operationalize usefulness, we review education and NLP work (Haladyna et al., 2002; Balepur et al., 2025), yielding three metrics:

1. **Difficulty:** How well the MCQ challenges test-takers (Akhtar et al., 2026, e.g., accuracy).
2. **Discriminability:** How well the MCQ splits ranks of low/high-skill test-takers, informatively discerning their skill (Polo et al., 2024).
3. **Quality:** How well the MCQ probes abilities without confusing test-takers, avoiding writing flaws (Moore et al., 2022, e.g., ambiguity).

If LLM-written distractors surpass human-guided distractors across all three metrics, LLMs could produce harder and higher-quality items than humans—justifying their use in MCQA benchmark construction. We discuss the metric implementations next.

#### 3.4.1 Difficulty and Discriminability

We score difficulty and discriminability with Item Response Theory (IRT; Lord and Novick, 2008). IRT estimates a test-taker's skill from its responses, then uses such skill to infer per-item difficulty and discriminability scores. Difficulty tests how hard an item is after accounting for test-taker skill, while discriminability tests how well an item distinguishes test-takers of varied skill levels. Both metrics are desirable; for example, if all models' accuracy is 25% on a benchmark, the benchmark is less useful as it cannot distinguish between models, even though the benchmark may appear "hard".

We implement a 3PL IRT model, which models the probability that a test-taker $j$ answers a multiple-choice item $i$ correctly $\Pr(Y_{ij} = 1)$, as a function of the item's difficulty $b_i$, discriminability $a_i$, and the probability of guessing correctly $c_i$:

$$\Pr(Y_{ij} = 1) = c_i + (1 - c_i) \times \sigma(1.702\, a_i(\theta_j - b_i))\,. \quad (1)$$

Intuitively, a test-taker whose ability $\theta_j$ matches the item's difficulty $b_i$ has a 50% chance of answering $i$ correctly, excluding guessing. Discriminability $a_i$ defines how sharply this probability increases with ability, while the guessing parameter $c_i$ defines how much guessing contributes to the probability of a correct response. $c_i$ has a prior centered at $\frac{1}{n}$ for $n$-choice questions. The 1.702 scaling factor is a standard implementation detail (Camilli, 1994). Appendix A.7 has the full implementation details.

To ensure our results are not reliant on our specific IRT model, we refit the model 18 times while ablating its design choices, such as holding out each of the 14 test-takers or excluding web-flagged ARC stems (Balepur et al., 2026). Trends across D-GEN often remain consistent (Appendix Table 11). Further, neither leaked answers nor length cues explain the D-GEN difficulty gains (Appendix A.6).

#### 3.4.2 Writing Quality

We test writing quality with BenchMarker (Balepur et al., 2026), a tool that uses LLM judges to flag 19 writing flaws from education research that mislead students (Tarrant et al., 2006), like ambiguity, implausible distractors, and items with more than one correct answer (full list in Appendix A.9). Balepur et al. (2026) validate BenchMarker with human annotations, and Moore et al. (2023) find that LLMs can reliably detect item-writing flaws on this rubric family; we use the exact same setup as the authors (details in Appendix A.9). We report the number of writing flaws per item as a writing quality metric.

## 4 Results: Are Harder LLM Distractors Better than Human-Guided Ones?

We now examine: are harder LLM distractors better than human-guided ones? We first show LLM distractors often make items harder, but these items are rarely more discriminative and can lower quality, so no LLM distractor sets beat human-guided ones across all three metrics (§4.1). We then show extra distractors can artificially drop scores without better separating LLM ranks (§4.2)—with difficulty possibly just stemming from selection bias versus the abilities MCQA intends to test (§4.3).

### 4.1 Harder Distractors Aren't Always Better

We test how LLM distractors impact MCQA evaluation, studying differences in our metrics (§3.4) over LLM-only and human-guided distractors (Figure 3).

LLM distractors only surpass human-guided distractors if they are harder, while also increasing discriminability and having fewer writing flaws (§3.4)—shown as a blue point in the shaded quadrant (Figure 3). Only two data points meet these criteria: Gemini on MMLU in D-GEN and GPT on MMLU in D-EXT. Yet neither is significantly more discriminative (Appendix Figure 7), so their gains could just be noise. Thus, no LLM distractor configuration significantly beats its human-guided counterpart across all three metrics. We discuss our findings from the two task settings in depth next.

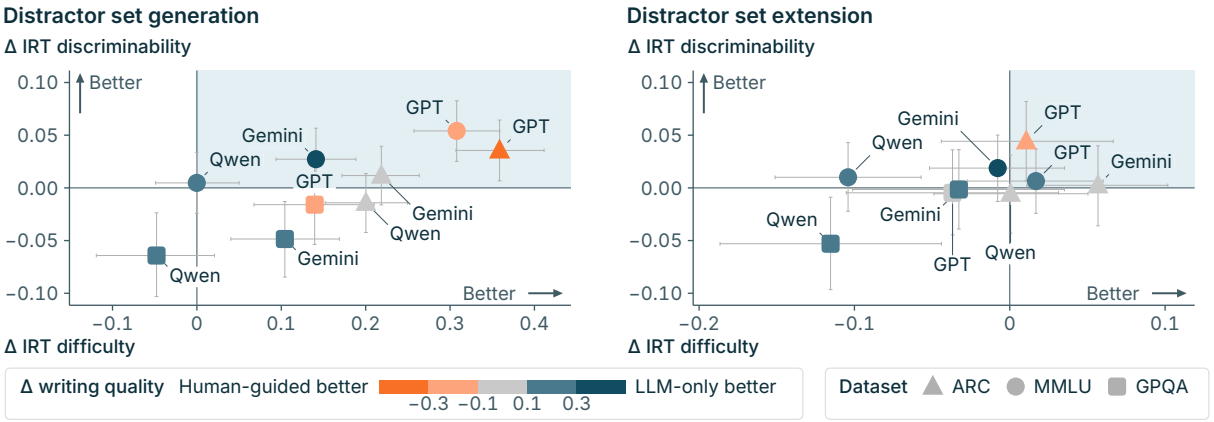



Figure 3: Differences between LLM and human-guided distractors across three metrics: difficulty, discriminability, and writing flaws (color). Negative differences mean that human-guided distractors surpass LLM distractors in that metric; error bars show 95% confidence intervals from a paired bootstrap over question stems. **GPT-based distractors are the most difficult, in part due to having the most writing flaws (orange).** Overall, LLM distractors are not significantly better than their human-guided counterparts on all three metrics (details in Appendix Figure 7).

**Distractor Generation.** 7/9 LLM distractor sets are significantly harder than human-guided items. GPT-5.2 heightens difficulty by 0.36, 0.31, and 0.14 on ARC, MMLU, and GPQA, but only one of these is significantly more discriminative (GPT on MMLU), and differences stay within $\pm 0.07$, so these harder items are rarely more discriminative. GPT, the hardest generator on every dataset, adds writing flaws on all three datasets (e.g., 0.6 per item on ARC), while Gemini and Qwen have fewer flaws than human-guided items on MMLU (0.41 and 0.29 fewer per item). Flaws explain part of GPT's difficulty (Moore et al., 2022): at equal flaws, its gains shrink by 13–20%, mostly due to items with multiple best answers (Appendix A.10).

**Distractor Extension.** Human-guided and model-only MCQs from D-EXT have few differences. Qwen distractors on GPQA and MMLU are easier ($-0.12$, $-0.10$). Extension in both settings increases difficulty versus human-guided D-GEN items ($+0.61$, $+0.60$), reducing the LLM-only difficulty benefits seen in D-GEN ($+0.14$). Differences between LLM-only and human-guided distractors are minimal in D-EXT, as both settings have more choices and rely on LLM generations. LLM-only extensions have fewer writing flaws in 4 of 9 cases, mirroring the same trend in distractor generation.

## 4.2 Extra Distractors Can Lower Accuracy Without Better Separating Rankings

D-EXT is often used to make benchmarks harder, as in MMLU-Pro (Wang et al., 2024) and MMMU-Pro (Yue et al., 2025), even scaling to an extreme 100 choices (Lee and Son, 2026). We test how this shifts evaluation by using GPT-5.2 to add $m \in [3, 6]$ distractors in D-EXT for the first 200 questions in ARC and MMLU. Scaling $m$ increases difficulty on ARC but fails to keep increasing discriminability on both datasets (Figure 4), suggesting the extra distractors artificially lower model scores without revealing new differences in LLM abilities. This further supports our main argument—harder benchmarks are not always better—and parallels insights in education: one high-quality distractor can be more informative than several weak ones (Papenberg and Musch, 2017).

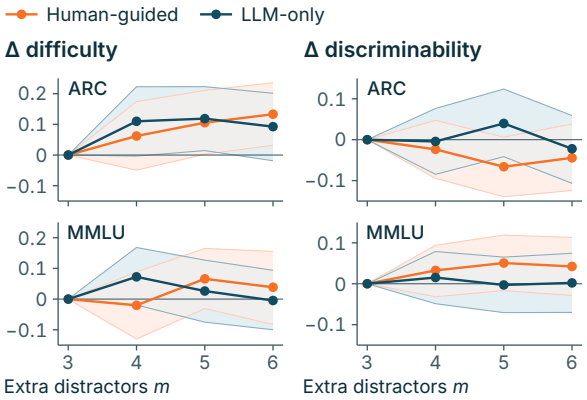



Figure 4: Change from $m = 3$ in difficulty and discriminability in D-EXT on ARC (top) and MMLU (bottom), using GPT-5.2 on the first 200 questions of ARC and MMLU each and all 14 test-taker models (95% CIs from a paired bootstrap test). Adding distractors can increase difficulty, but it barely helps to separate model ranks.

## 4.3 The Difficulty of Extra Distractors May Be Partially Driven by Selection Bias

To further understand the drivers of difficulty in D-EXT, we analyze the distribution of test-taker

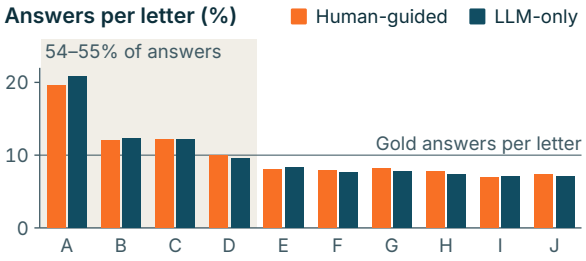



Figure 5: Distribution of test-taker responses across answer letters in D-EXT; the line marks 10%, the share of gold answers expected at each letter (observed: 9.3–10.8%). Test-taker LLMs pick A–D in 54–55% of cases, but these have just 40% of gold answers, for human-guided and LLM-only items. Thus, D-EXT's difficulty may partly stem from models' preference toward earlier answer letters versus abilities MCQA aims to assess.

responses across answer letters, as ten-option items place 60% of gold answers in E–J, letters that four-choice MCQs never use. Test-taker LLMs show a preference for early answer choice letters, replicating prior work (Zheng et al., 2024; Pezeshkpour and Hruschka, 2024). In D-EXT, they choose A–D in 54–55% of responses, but these contain just 40% of gold answers, for both human-guided and LLM-only items (Figure 5). We speculate this could stem from LLM training data mostly including MCQA examples with answers in A–D, or failures in long-context comprehension (Karpinska et al., 2024).

Lee and Son (2026) also show that models suffer from position bias toward early options (A–D) under uncertainty in larger MCQA answer choice sets, suggesting that this is a general behavior. Overall, these results are further evidence suggesting extension may make MCQs harder by targeting shallow positional biases or long-context limitations, rather than probing abilities that MCQA intends to test like knowledge recall and reasoning (Krathwohl, 2002).

## 5 Analysis: Why Does Harder ≠ Better?

Section 4 showed that LLM distractors often make items harder without making them better; we now analyze the distractors to help explain why. After showing our LLM distractors are not just recycled (§5.1), we explore types of writing flaws they add (§5.2) and how LLMs construct distractors (§5.3).

### 5.1 LLM Distractors Are Not Just Recycled

As we rely on existing MCQA benchmarks, perhaps LLMs sometimes match the usefulness of human-guided items via contamination—recalling the existing item (Deng et al., 2024). To test this, we see how often LLM-only distractors in D-GEN match any distractor in the original benchmark. Following Du et al. (2025), we detect this with a 0.90 cosine similarity threshold via MiniLM-L6-v2 (Reimers and Gurevych, 2019) for textual distractors, and exact match for numeric distractors.

| | GPT-5.2 | Gemini-3.1 Pro | Qwen-3.5 397B |
|---|---|---|---|
| ARC | 14.7% | 22.2% | 19.8% |
| MMLU | 8.3% | 15.8% | 12.3% |
| GPQA | 19.4% | 24.8% | 21.4% |

Table 1: LLM-only distractors in D-GEN that match an original benchmark distractor (0.90 cosine similarity for text-based distractors, exact match for numeric ones). Most LLM-only distractors do not match the benchmark distractor, meaning these items are not just memorized.

| | D-GEN | | D-EXT | |
|---|---|---|---|---|
| Writing Flaw Type | Human | LLM | Human | LLM |
| Unequal Option Length | 20.4 | 28.0 | 37.5 | 37.0 |
| Implausible Distractors | 11.2 | 7.8 | 22.4 | 19.3 |
| Has Absolute Terms | 9.6 | 14.4 | 23.2 | 23.7 |
| Multiple Best Answers | 12.2 | 15.3 | 29.3 | 28.6 |
| Has Convergence Cues | 21.6 | 19.1 | 20.6 | 19.3 |

Table 2: Percentage of five writing flaws in human-guided and LLM-only distractors in D-GEN and D-EXT, averaged across generator models. LLMs (mostly GPT) struggle to maintain option length and limit plausible answers. Appendix A.9 has all 19 flaws.

Exact overlaps are sparse (Table 1), and this overlap is less than overlap between two generator models on the same stem: 26% vs. 19% (ARC), 18% vs. 12% (MMLU), 33% vs. 22% (GPQA). Thus, LLMs recover the original distractors less often than they agree with each other. On GPQA, 80% of matches are exact matches on numeric distractors, so exact matches are likely a result of limited answer pools. Appendix Table 11 further shows D-GEN difficulty gains hold when excluding ARC stems detected as contaminated via web search (Li et al., 2024). While these analyses may not flag all overlap types (e.g., knowledge overlaps), they detect the simplest ones and gives us confidence that LLMs can generate distractors that are not purely recycled.

### 5.2 Rife Writing Flaws in LLM Distractors

To further study LLM-only and human-guided distractors, we compare their distributions of violations across the 19 writing flaw types from §3.4.

In D-GEN, LLM distractors have mixed effects on writing quality over human-guided items (Table 2), and most differences are model specific (Appendix Table 16). LLMs add fewer implausible distractors, but GPT adds long, obvious distractors (unequal option length +27.7 points over its human copy), and GPT and Qwen add more "multiple best answers" (+7.7, +1.8), suggesting they overcom-

| Model | SOLVE | SIMULATE | RECALL | SHALLOW | REFINE |
|---|---|---|---|---|---|
| GPT | 9% | 46% | 31% | 76% | 6% |
| Gemini | 56% | 66% | 40% | 58% | 16% |
| Qwen | 6% | 89% | 70% | 36% | 51% |

Table 3: Traces can use many strategies; except SHALLOW, educators use similar strategies (Gierl et al., 2017). How often LLMs use varied reasoning trace strategies when making 20 GPQA distractor sets per generator in D-GEN, averaged over four annotators. Appendix A.11 has definitions, per-strategy agreement, and an example.

pensate, struggling to weigh plausibility (Lee et al., 2025). Figure 1 shows examples: 10/14 test-takers pick a GPT distractor that is plausibly correct, and no test-taker picks GPT's long, obvious distractors.

In D-EXT, differences are tempered, as writing flaws increase over all distractor sets. This complements our analysis in §4.2: extra distractors can scale difficulty, but also at the expense of increased writing flaws. Overall, we advise NLP researchers to apply auditing tools like LLM judges (§3.4) to LLM distractors, as they can find areas to intervene (e.g., shortening lengthy distractors) and transparently document where difficulty could stem from.

### 5.3 How Do LLMs Construct Distractors?

We lastly see *how* LLMs create distractors via their reasoning traces.[3] Inspired by Bingham (2023), two authors first reviewed 60 traces to derive five strategies: 1) SOLVE the item, 2) SIMULATE test-taker mistakes, 3) RECALL relevant facts, 4) add a SHALLOW perturbation of the gold answer (e.g., negative sign), and 5) REFINE distractors. Four annotators then label 20 GPQA D-GEN traces per generator with these strategies with Fleiss' $\kappa = 0.66$ (Fleiss, 1971)—substantial agreement (Landis and Koch, 1977, details in Appendix A.11).

At a high level, LLMs follow similar strategies to educators, barring SHALLOW, such as re-solving the item, recalling related concepts, simulating mistakes, and refining items (Gierl et al., 2017). For example, in one case, GPT finds a common misconception in RECALL: "*I'll consider $CO_2$ as there's a common misconception that plants release it.*"

While promising, the most common strategies for GPT and Gemini are SHALLOW perturbations (76% and 58% of traces on average; Table 3)—a strategy educators are discouraged from using (Haladyna et al., 2002). Further, while many LLMs attempt SIMULATE, they do not always aptly target test-taker errors; they often perturb steps in their own solution versus focusing on high-level mistakes. To illustrate, on one MCQ where test-takers need to solve a physics equation with the answer 3pc, Gemini's distractors drop or invert a correction/temperature term (Appendix Trace A.1). On another MCQ on quantum expectation value, human distractors follow common calculation errors (e.g., skip a normalization step), while Qwen's distractors generate derivatives of the gold answer ($27/5\ E_0$ and $5/2\ E_0$ for $81/23\ E_0$). While technically valid, PhD-level test-takers (the expected skill level of GPQA) rarely make such simple errors.

We advise researchers to study generator reasoning traces when creating distractors, ensuring they target errors calibrated to their desired skill level.

## 6 Conclusion: Tips for LLM Distractors

We extensively probe LLM-generated distractors for NLP evaluation via item response theory, writing quality detection, contamination checks, and qualitative analyses. We find LLM distractors often make items harder, but these harder items are rarely more discriminative and can have lower quality, so lower model accuracy cannot conclusively show that LLM-generated distractors are useful. We distill our results into six steps for benchmark design:

1. Use humans to author a subset of distractors.
2. Use our evaluation framework to score human-written and LLM distractors across difficulty, discriminability, and writing flaws (§3).
3. Do not focus on lower accuracy alone for distractor usefulness; understand trade-offs of all three metrics to decide which generator model best optimizes the metrics you value (§4.1).
4. Decide whether extension is needed; check whether it solely lowers scores and reflects position bias (§4.3) or also improves discrimination between model rankings (§4.2).
5. Analyze distractors models pick to understand common errors, and find never-chosen options that may be removed for efficiency (Figure 1).
6. Qualitatively analyze items, such as writing flaw breakdowns (§5.2) and reasoning trace analyses (§5.3) to find areas for improvement and human intervention.

Creating MCQA benchmarks is tedious, but necessary to make progress in NLP (Clark et al., 2020). Our results show LLMs can help with these efforts via distractor generation, but evaluation requires metrics beyond the accuracy of test-taker models.

[3] GPT and Gemini return distilled/summarized reasoning traces, but we still find them informative for this analysis.

## 7 Limitations

LLMs are sensitive to prompting (Alzahrani et al., 2024), so modifying our prompts for distractor generation and extension (§3.3) and MCQA evaluation (§3.3) could impact our results. For controlled comparisons, we run all test-taker models on a standardized prompt, the de-facto evaluation approach for our datasets (ARC, MMLU, and GPQA). Improvements to prompt design could change the difficulty and writing quality trade-offs we report. Future work can also paraphrase distractors to see if they stay effective, or add a prompt clause like "Some options may be distractors. Read each carefully before answering." Appendix A.3 has our prompts for reproducibility.

Our IRT model assumes that each test-taker has a fixed skill, but this is a standard modeling choice when using IRT in both education and NLP research (§2, §3.4). A random sample of the at-large "LLM population" is not possible as there is no "true" sample of LLMs, so we follow standardized procedures from Rodriguez et al. (2021) when using IRT in NLP. We select a variety of models, 1B–32B open-weight models and an API model, to calculate a well-spread ability scale (Appendix Table 8, where ability ranges from $+2.59$ to $-1.59$). Due to computational constraints, our test-taker pool skews toward smaller open-weight models: only two of the 14 test-takers have ability above $+0.6$ (Appendix Table 8), so estimates for the hardest items, especially for GPQA questions, depend on relatively few test-taker responses; in the future, adding stronger test-takers would increase accuracy of estimates. We study distractor count (§4.2) with only GPT-5.2 on the first 200 questions of ARC and MMLU each; cell means have standard errors of about 0.09 from stem variance, but paired difficulty differences have 95% confidence intervals of about $\pm 0.11$ per dataset, which future work can narrow by using all stems.

Further, we cannot eliminate the possibility of generation or test-taker models having already seen the human MCQs without access to their training data (Soldaini et al., 2024). All experiment setups use the same question stem and gold answer, and only change the source of distractor choices (human-guided versus LLM-generated), so any gains from memorization of the stem and answer are identical across conditions; however, a test-taker that has memorized a full human MCQ could still find human-guided items easier. As a check, we use BenchMarker web-search flags (Balepur et al., 2026) on ARC: $\sim$40% of stems are on the web, but D-GEN difficulty gains hold on questions not flagged by web search (§3.4). Thus, we encourage future work to adopt our evaluation setup on their own (and presumably unseen) MCQA benchmarks. Such evaluations would also strengthen the generalization of our analyses. We open-source all code/data to support these efforts.

Finally, we acknowledge many auxiliary dimensions that we could focus our evaluation on (such as generation model costs and studying inter-model versus intra-model extension) that we did not consider due to resource constraints. We specifically target the question of the role of human guidance in MCQA benchmark design with three closed and open-weight generator models, a useful prerequisite before exploring these follow-up analyses. We release this paper to inspire such future work.

## 8 Ethical Considerations

While we show the benefits of using LLMs to support MCQA benchmark design, we never argue that these items are sufficient for real student testing. In fact, §5.3 reveals issues in LLM-generated MCQs that may degrade usefulness in education settings, so we urge researchers to more rigorously evaluate MCQs before using them with real students. Further, there exist other instantiations of MCQ quality for NLP evaluation that our paper only partly considers (§4.2; Appendix A.6), including shortcuts (Lai et al., 2021), biases (Wei et al., 2024), and self-preference (Panickssery et al., 2024), which may be worth measuring before deciding to release LLM-generated MCQA benchmarks.

Generative AI (GenAI) was used in this project. We used Codex[4] to design plots and refactor code, and GPT-5 to refine paper writing for brevity. We never use GenAI for writing text from scratch in this paper. We take complete responsibility for any GenAI errors. By discussing GenAI usage here, we aim to encourage other researchers to do the same.

## References

Mubashara Akhtar, Anka Reuel, Prajna Soni, Sanchit Ahuja, Pawan Sasanka Ammanamanchi, Ruchit Rawal, Vilém Zouhar, Srishti Yadav, Chenxi Whitehouse, Dayeon Ki, Jennifer Mickel, Leshem Choshen, Marek Šuppa, Jan Batzner, Jenny Chim, Jeba Sania, Yanan Long, Hossein A. Rahmani, Christina Knight,

[4] https://openai.com/codex/

and 18 others. 2026. When ai benchmarks plateau: A systematic study of benchmark saturation. *Preprint*, arXiv:2602.16763.

Syeda Nahida Akter, Shrimai Prabhumoye, Matvei Novikov, Seungju Han, Ying Lin, Evelina Bakhturina, Eric Nyberg, Yejin Choi, Mostofa Patwary, Mohammad Shoeybi, and 1 others. 2026. Nemotron-crossthink: Scaling self-learning beyond math reasoning. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 984–1002.

Elaf Alhazmi, Quan Z. Sheng, Wei Emma Zhang, Munazza Zaib, and Ahoud Alhazmi. 2024. Distractor generation in multiple-choice tasks: A survey of methods, datasets, and evaluation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 14437–14458, Miami, Florida, USA. Association for Computational Linguistics.

Norah Alzahrani, Hisham Alyahya, Yazeed Alnumay, Sultan AlRashed, Shaykhah Alsubaie, Yousef Almushayqih, Faisal Mirza, Nouf Alotaibi, Nora Al-Twairesh, Areeb Alowisheq, M Saiful Bari, and Haidar Khan. 2024. When benchmarks are targets: Revealing the sensitivity of large language model leaderboards. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13787–13805, Bangkok, Thailand. Association for Computational Linguistics.

Alexander Amini, Anna Banaszak, Harold Benoit, Arthur Böök, Tarek Dakhran, Song Duong, Alfred Eng, Fernando Fernandes, Marc Härkönen, Anne Harrington, and 1 others. 2025. Lfm2 technical report. *arXiv preprint arXiv:2511.23404*.

Kyunghoon Bae, Eunbi Choi, Kibong Choi, Stanley Jungkyu Choi, Yemuk Choi, Kyubeen Han, Seokhee Hong, Junwon Hwang, Taewan Hwang, Joonwon Jang, and 1 others. 2025. Exaone 4.0: Unified large language models integrating non-reasoning and reasoning modes. *arXiv preprint arXiv:2507.11407*.

Nishant Balepur, Bhavya Rajasekaran, Jane Oh, Michael Xie, Atrey Desai, Vipul Gupta, Steven James Moore, Eunsol Choi, Rachel Rudinger, and Jordan Lee Boyd-Graber. 2026. Benchmarker: An education-inspired toolkit for highlighting flaws in multiple-choice benchmarks. *arXiv preprint arXiv:2602.06221*.

Nishant Balepur, Abhilasha Ravichander, and Rachel Rudinger. 2024. Artifacts or abduction: How do llms answer multiple-choice questions without the question? In *Annual Meeting of the Association for Computational Linguistics*.

Nishant Balepur, Rachel Rudinger, and Jordan Lee Boyd-Graber. 2025. Which of these best describes multiple choice evaluation with LLMs? a) forced B) flawed C) fixable D) all of the above. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3394–3418, Vienna, Austria. Association for Computational Linguistics.

Aarti Basant, Abhijit Khairnar, Abhijit Paithankar, Abhinav Khattar, Adithya Renduchintala, Aditya Malte, Akhiad Bercovich, Akshay Hazare, Alejandra Rico, Aleksander Ficek, and 1 others. 2025. Nvidia nemotron nano 2: An accurate and efficient hybrid mamba-transformer reasoning model. *arXiv preprint arXiv:2508.14444*.

Stella Biderman, Hailey Schoelkopf, Lintang Sutawika, and 1 others. 2024. Lessons from the trenches on reproducible evaluation of language models. *arXiv preprint arXiv:2405.14782*.

Andrea J Bingham. 2023. From data management to actionable findings: A five-phase process of qualitative data analysis. *International journal of qualitative methods*, 22:16094069231183620.

Semere Kiros Bitew, Johannes Deleu, Chris Develder, and Thomas Demeester. 2023. Distractor generation for multiple-choice questions with predictive prompting and large language models. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, pages 48–63. Springer.

Grace Byun and Jinho D. Choi. 2025. *D-GEN*: Automatic distractor generation and evaluation for reliable assessment of generative models. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 3316–3349, Vienna, Austria. Association for Computational Linguistics.

Gregory Camilli. 1994. Teacher's corner: origin of the scaling constant d= 1.7 in item response theory. *Journal of educational and behavioral statistics*, 19(3):293–295.

Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. Think you have solved question answering? try arc, the ai2 reasoning challenge. *arXiv preprint arXiv:1803.05457*.

Peter Clark, Oren Etzioni, Tushar Khot, Daniel Khashabi, Bhavana Mishra, Kyle Richardson, Ashish Sabharwal, Carissa Schoenick, Oyvind Tafjord, Niket Tandon, and 1 others. 2020. From 'f'to 'a'on the ny regents science exams: An overview of the aristo project. *Ai Magazine*, 41(4):39–53.

Chunyuan Deng, Yilun Zhao, Xiangru Tang, Mark Gerstein, and Arman Cohan. 2024. Investigating data contamination in modern benchmarks for large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 8706–8719.

Xeron Du, Yifan Yao, Kaijing Ma, Bingli Wang, Tianyu Zheng, King Zhu, Minghao Liu, Yiming Liang, Xiaolong Jin, Zhenlin Wei, Chujie Zheng, Kaixin Deng, Shuyue Guo, Shian Jia, Sichao Jiang, Yiyan Liao, Rui Li, Qinrui Li, Sirun Li, and 76 others. 2025. SuperGPQA: Scaling LLM evaluation across 285 graduate disciplines. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.

Wanyong Feng, Jaewook Lee, Hunter McNichols, Alexander Scarlatos, Digory Smith, Simon Woodhead, Nancy Otero Ornelas, and Andrew Lan. 2024. Exploring automated distractor generation for math multiple-choice questions via large language models. *arXiv preprint arXiv:2404.02124*.

Joseph L. Fleiss. 1971. Measuring nominal scale agreement among many raters. *Psychological Bulletin*, 76(5):378–382.

Mark J Gierl, Okan Bulut, Qi Guo, and Xinxin Zhang. 2017. Developing, analyzing, and using distractors for multiple-choice tests in education: A comprehensive review. *Review of educational research*, 87(6):1082–1116.

Google DeepMind. 2026. Gemini 3.1 Pro model card. https://deepmind.google/models/model-cards/gemini-3-1-pro/. Accessed: 2026-05-17.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, and 1 others. 2024. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Yuling Gu, Oyvind Tafjord, Bailey Kuehl, Dany Haddad, Jesse Dodge, and Hannaneh Hajishirzi. 2025. OLMES: A standard for language model evaluations. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 5005–5033, Albuquerque, New Mexico. Association for Computational Linguistics.

Thomas M Haladyna and Steven M Downing. 1989. A taxonomy of multiple-choice item-writing rules. *Applied measurement in education*, 2(1):37–50.

Thomas M Haladyna, Steven M Downing, and Michael C Rodriguez. 2002. A review of multiple-choice item-writing guidelines for classroom assessment. *Applied measurement in education*, 15(3):309–333.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021. Measuring massive multitask language understanding. In *International Conference on Learning Representations*.

Valentin Hofmann, David Heineman, Ian Magnusson, Kyle Lo, Jesse Dodge, Maarten Sap, Pang Wei Koh, Chun Wang, Hannaneh Hajishirzi, and Noah A. Smith. 2025. Fluid language model benchmarking. In *Second Conference on Language Modeling*.

IBM Research. 2025. Granite 4.0 language models. https://github.com/ibm-granite/granite-4.0-language-models. Accessed: 2025-10-01.

Marzena Karpinska, Katherine Thai, Kyle Lo, Tanya Goyal, and Mohit Iyyer. 2024. One thousand and one pairs: A" novel" challenge for long-context language models. *arXiv preprint arXiv:2406.16264*.

Alex Kipnis, Konstantinos Voudouris, Luca M. Schulze Buschoff, and Eric Schulz. 2024. metabench – a sparse benchmark of reasoning and knowledge in large language models. *arXiv preprint arXiv:2407.12844*.

DR Krathwohl. 2002. A revision bloom's taxonomy: An overview. *Theory into Practice*.

Hollis Lai, Mark J Gierl, Claire Touchie, Debra Pugh, André-Philippe Boulais, and André De Champlain. 2016. Using automatic item generation to improve the quality of mcq distractors. *Teaching and learning in medicine*, 28(2):166–173.

Yuxuan Lai, Chen Zhang, Yansong Feng, Quzhe Huang, and Dongyan Zhao. 2021. Why machine reading comprehension models learn shortcuts? *arXiv preprint arXiv:2106.01024*.

John P. Lalor, Hao Wu, and Hong Yu. 2016. Building an evaluation scale using item response theory. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*. ArXiv:1605.08889.

J Richard Landis and Gary G Koch. 1977. The measurement of observer agreement for categorical data. *biometrics*, pages 159–174.

Nahyun Lee and Guijin Son. 2026. Pushing the boundaries of multiple choice evaluation to one hundred options. *arXiv preprint arXiv:2604.14634*.

Yooseop Lee, Suin Kim, and Yohan Jo. 2025. Generating plausible distractors for multiple-choice questions via student choice prediction. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 23669–23692, Vienna, Austria. Association for Computational Linguistics.

Ming Li, Pei Chen, Chenguang Wang, Hongyu Zhao, Yijun Liang, YuPeng Hou, Fuxiao Liu, and Tianyi Zhou. 2025a. Mosaic-IT: Cost-free compositional data synthesis for instruction tuning. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 25287–25318, Vienna, Austria. Association for Computational Linguistics.

Xiang Lisa Li, Farzaan Kaiyom, Evan Zheran Liu, Yifan Mai, Percy Liang, and Tatsunori Hashimoto. 2025b. Autobencher: Towards declarative benchmark construction. In *The Thirteenth International Conference on Learning Representations*.

Yucheng Li, Yunhao Guo, Frank Guerin, and Chenghua Lin. 2024. An open-source data contamination report for large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 528–541, Miami, Florida, USA. Association for Computational Linguistics.

Chen Liang, Xiao Yang, Neisarg Dave, Drew Wham, Bart Pursel, and C Lee Giles. 2018. Distractor generation for multiple choice questions using learning to rank. In *Proceedings of the thirteenth workshop on innovative use of NLP for building educational applications*, pages 284–290.

Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, Benjamin Newman, Binhang Yuan, Bobby Yan, Ce Zhang, Christian Cosgrove, Christopher D Manning, Christopher Re, Diana Acosta-Navas, Drew A. Hudson, and 31 others. 2023. Holistic evaluation of language models. *Transactions on Machine Learning Research*. Featured Certification, Expert Certification, Outstanding Certification.

Alisa Liu, Swabha Swayamdipta, Noah A Smith, and Yejin Choi. 2022. Wanli: Worker and ai collaboration for natural language inference dataset creation. *arXiv preprint arXiv:2201.05955*.

Ruibo Liu, Jerry Wei, Fangyu Liu, Chenglei Si, Yanzhe Zhang, Jinmeng Rao, Steven Zheng, Daiyi Peng, Diyi Yang, Denny Zhou, and Andrew M. Dai. 2024. Best practices and lessons learned on synthetic data. In *First Conference on Language Modeling*.

Lin Long, Rui Wang, Ruixuan Xiao, Junbo Zhao, Xiao Ding, Gang Chen, and Haobo Wang. 2024. On LLMs-driven synthetic data generation, curation, and evaluation: A survey. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 11065–11082, Bangkok, Thailand. Association for Computational Linguistics.

Frederic M Lord and Melvin R Novick. 2008. *Statistical theories of mental test scores*. IAP.

Inbal Magar and Roy Schwartz. 2022. Data contamination: From memorization to exploitation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 157–165, Dublin, Ireland. Association for Computational Linguistics.

Ruslan Mitkov and 1 others. 2003. Computer-aided generation of multiple-choice tests. In *Proceedings of the HLT-NAACL 03 workshop on Building educational applications using natural language processing*, pages 17–22.

Steven Moore, Huy A Nguyen, Norman Bier, Tanvi Domadia, and John Stamper. 2022. Assessing the quality of student-generated short answer questions using gpt-3. In *European conference on technology enhanced learning*, pages 243–257. Springer.

Steven Moore, Huy A Nguyen, Tianying Chen, and John Stamper. 2023. Assessing the quality of multiple-choice questions using gpt-4 and rule-based methods. In *European Conference on Technology Enhanced Learning*, pages 229–245. Springer.

Sérgio Silva Mucciaccia, Thiago Meireles Paixão, Filipe Wall Mutz, Claudine Santos Badue, Alberto Ferreira de Souza, and Thiago Oliveira-Santos. 2025. Automatic multiple-choice question generation and evaluation systems based on llm: A study case with university resolutions. In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 2246–2260.

Mihai Nadăș, Laura Dioșan, and Andreea Tomescu. 2025. Synthetic data generation using large language models: Advances in text and code. *IEEE Access*.

Team Olmo, Allyson Ettinger, Amanda Bertsch, Bailey Kuehl, David Graham, David Heineman, Dirk Groeneveld, Faeze Brahman, Finbarr Timbers, Hamish Ivison, and 1 others. 2025. Olmo 3. *arXiv preprint arXiv:2512.13961*.

OpenAI. 2025. Introducing GPT-5.2. https://openai.com/index/introducing-gpt-5-2/. Accessed: 2026-05-17.

Arjun Panickssery, Samuel R. Bowman, and Shi Feng. 2024. LLM evaluators recognize and favor their own generations. In *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*.

Martin Papenberg and Jochen Musch. 2017. Of small beauties and large beasts: The quality of distractors on multiple-choice tests is more important than their quantity. *Applied Measurement in Education*, 30(4):273–286.

Pouya Pezeshkpour and Estevam Hruschka. 2024. Large language models sensitivity to the order of options in multiple-choice questions. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 2006–2017, Mexico City, Mexico. Association for Computational Linguistics.

Long Phan, Alice Gatti, Ziwen Han, Nathaniel Li, Josephina Hu, Hugh Zhang, Sean Shi, Michael Choi, Anish Agrawal, Arnav Chopra, and 1 others. 2025. Humanity's last exam. *arXiv preprint arXiv:2501.14249*.

Felipe Maia Polo, Lucas Weber, Leshem Choshen, Yuekai Sun, Gongjun Xu, and Mikhail Yurochkin. 2024. tinybenchmarks: evaluating llms with fewer examples. In *Proceedings of the 41st International Conference on Machine Learning*, ICML'24. JMLR.org.

Fanyi Qu, Hao Sun, and Yunfang Wu. 2024. Unsupervised distractor generation via large language model distilling and counterfactual contrastive decoding. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 827–838.

Qwen Team. 2026. Qwen3.5-397B-A17B. https://huggingface.co/Qwen/Qwen3.5-397B-A17B. Accessed: 2026-05-17.

Nils Reimers and Iryna Gurevych. 2019. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics.

David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R. Bowman. 2024. GPQA: A graduate-level google-proof q&a benchmark. In *First Conference on Language Modeling*.

Pedro Rodriguez, Joe Barrow, Alexander Miserlis Hoyle, John P. Lalor, Robin Jia, and Jordan Boyd-Graber. 2021. Evaluation examples are not equally informative: How should that change NLP leaderboards? In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 4486–4503, Online. Association for Computational Linguistics.

Robin Schmucker and Steven Moore. 2025. The impact of item-writing flaws on difficulty and discrimination in item response theory. *arXiv preprint arXiv:2503.10533*.

Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. 2024. Ai models collapse when trained on recursively generated data. *Nature*, 631(8022):755–759.

Aaditya Singh, Adam Fry, Adam Perelman, Adam Tart, Adi Ganesh, Ahmed El-Kishky, Aidan McLaughlin, Aiden Low, AJ Ostrow, Akhila Ananthram, and 1 others. 2025. Openai gpt-5 system card. *arXiv preprint arXiv:2601.03267*.

Luca Soldaini, Rodney Kinney, Akshita Bhagia, Dustin Schwenk, David Atkinson, Russell Authur, Ben Bogin, Khyathi Chandu, Jennifer Dumas, Yanai Elazar, Valentin Hofmann, Ananya Jha, Sachin Kumar, Li Lucy, Xinxi Lyu, Nathan Lambert, Ian Magnusson, Jacob Morrison, Niklas Muennighoff, and 17 others. 2024. Dolma: an open corpus of three trillion tokens for language model pretraining research. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15725–15788, Bangkok, Thailand. Association for Computational Linguistics.

Yoo Yeon Sung, Maharshi Gor, Eve Fleisig, Ishani Mondal, and Jordan Lee Boyd-Graber. 2025. Is your benchmark truly adversarial? AdvScore: Evaluating human-grounded adversarialness. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 623–642, Albuquerque, New Mexico. Association for Computational Linguistics.

Marie Tarrant, Aimee Knierim, Sasha K Hayes, and James Ware. 2006. The frequency of item writing flaws in multiple-choice questions used in high stakes nursing assessments. *Nurse Education Today*, 26(8):662–671.

Gemma Team, Aishwarya Kamath, Johan Ferret, Shreya Pathak, Nino Vieillard, Ramona Merhej, Sarah Perrin, Tatiana Matejovicova, Alexandre Ramé, Morgane Rivière, Louis Rouillard, Thomas Mesnard, Geoffrey Cideron, Jean bastien Grill, Sabela Ramos, Edouard Yvinec, Michelle Casbon, Etienne Pot, Ivo Penchev, and 197 others. 2025. Gemma 3 technical report. *Preprint*, arXiv:2503.19786.

UK AISI. 2024. Inspect AI: Framework for Large Language Model Evaluations.

Clara Vania, Phu Mon Htut, William Huang, Dhara Mungra, Richard Yuanzhe Pang, Jason Phang, Haokun Liu, Kyunghyun Cho, and Samuel R. Bowman. 2021. Comparing test sets with item response theory. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 1141–1158, Online. Association for Computational Linguistics.

Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyan Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhu Chen. 2024. MMLU-pro: A more robust and challenging multi-task language understanding benchmark. In *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.

Sheng-Lun Wei, Cheng-Kuang Wu, Hen-Hsen Huang, and Hsin-Hsi Chen. 2024. Unveiling selection biases: Exploring order and token sensitivity in large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 5598–5621, Bangkok, Thailand. Association for Computational Linguistics.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, and 1 others. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Xiang Yue, Tianyu Zheng, Yuansheng Ni, Yubo Wang, Kai Zhang, Shengbang Tong, Yuxuan Sun, Botao Yu, Ge Zhang, Huan Sun, Yu Su, Wenhu Chen, and Graham Neubig. 2025. MMMU-pro: A more robust multi-discipline multimodal understanding benchmark. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*.

Rowan Zellers, Yonatan Bisk, Roy Schwartz, and Yejin Choi. 2018. SWAG: A large-scale adversarial dataset for grounded commonsense inference. In *Proceedings of the 2018 Conference on Empirical Methods in*

*Natural Language Processing*, pages 93–104, Brussels, Belgium. Association for Computational Linguistics.

Lishan Zhang and Kurt VanLehn. 2021. Evaluation of auto-generated distractors in multiple choice questions from a semantic network. *Interactive Learning Environments*, 29(6):1019–1036.

Yuhui Zhang, Yuchang Su, Yiming Liu, Xiaohan Wang, James Burgess, Elaine Sui, Chenyu Wang, Josiah Aklilu, Alejandro Lozano, Anjiang Wei, and 1 others. 2025. Automated generation of challenging multiple-choice questions for vision language model evaluation. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 29580–29590.

Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. 2024. Large language models are not robust multiple choice selectors. In *The Twelfth International Conference on Learning Representations*.

# A Appendix

## A.1 Data

We initially sample 1000 random examples from the test set of ARC-Challenge (Clark et al., 2018) and MMLU (Hendrycks et al., 2021). MMLU questions were drawn from a pool of question seeds used in MMLU-Pro (Wang et al., 2024). We use 448 examples of GPQA (Rein et al., 2024), the full question set. All datasets are publicly available, so our experiments are within their intended use. ARC-Challenge and GPQA are licensed under CC BY 4.0 and MMLU under MIT. We did not collect any datasets, so we did not check for PII. To our knowledge, all questions are in English.

We check for bad questions using automated methods (check for missing question stems and answer choices) and manual inspection. We filter out one MMLU question (QA.1) and four GPQA questions (QA.2, QA.3, QA.4, QA.5).

Additionally, we filter questions if a generator model creating MCQs fails to produce a valid output across any setting for a given human MCQ seed, yielding **995** total items in ARC-Challenge, **954** in MMLU, and **415** in GPQA.

We use datasets consistent with their intended use in NLP research and evaluating model knowledge and reasoning capabilities. We use this data as seeds for downstream question generation. Any generated data released with this paper should not be used unsupervised or as replacements for educational tests without additional review as mentioned in recommended guidelines in Section 6.

We do not collect new human-subject data. We manually inspect a portion of the generated artifacts (stems, answer options, generated distractors, model traces) and to our best knowledge, we do not find PII or offensive content.

## A.2 InspectAI Implementation

InspectAI[5] is a recent effort from the United Kingdom's AI Security Institute to standardize NLP evaluations (UK AISI, 2024). Additionally, we also use InspectAI for initial creation of LLM-generated questions, as described below. Any InspectAI framework contains three parts:

1. **Task:** The data for the task. We have separate tasks for (1) question generation and (2) test-taker evaluation. For (1), each task is made up of a question stem, gold answer, and human distractor in the distractor set generation setting and required distractors from the prior setting for the distractor set extension setting. For (2), each task contains the question stem, randomized answer choices, the index of the gold answer, and associated metadata for data classification.

2. **Solver:** The NLP system that solves the task. For (1), given an QA pair or complete MCQ, the solver prompts a generator LLM to create three distractors or six distractors, respectively. Following generation, it checks JSON output for invalid or duplicate distractors, shuffles answer choices, and logs the question with associated metadata including model reasoning traces. For (2), the solver prompts a test-taker LLM with either a full question or choices-only prompt and parses for an answer letter in the generated JSON. If no valid answer letter is parsed, the solver logs a fallback and assigns a random answer letter.

3. **Scorer:** How task success/failure is evaluated. For (1) and (2), the scorer records if an item was successfully generated; additionally, the scorer for (2) computes item accuracy and records other information such as the gold answer letter.

Test-takers answer from a closed answer choice set (Prompt A.1), so hallucinations can only surface if the model fails to produce a valid answer or if the model selects an incorrect answer. For the former, across 496,440 test-taker evaluations, 98.84% of responses are parsable to a valid choice letter; for the remaining 1.16%, there is a logged fallback. For the latter, the difficulty metric is meant to account for incorrect answers.

## A.3 LLM Details

We access all models through InspectAI; models are passed through using provider-prefixed model identifiers. Hosted models use InspectAI provider strings such as `openai/...` for GPT models, `google/...` for Gemini models, and `together/...` for Qwen models. Open-weight models are locally run using `vllm/...` identifiers.

For generation, we use three generator LLMs:

- openai/gpt-5.2-2025-12-11
- google/gemini-3.1-pro-preview
- together/Qwen/Qwen3.5-397B-A17B

[5] https://inspect.aisi.org.uk/

For generation, we specify the following parameters: maximum generation length of 10,000 tokens, default temperature, two model-call retries if initial generation fails, and a maximum of three parsing attempts for each item. When supported by models, we use medium reasoning effort. This corresponds to medium in GPT-5.2 and Gemini-3.1 Pro, and goes unused for Qwen3.5-397B-A17B (not supported). We instead enable thinking mode through the binary toggle.

GPT-5.2 and Gemini-3.1 Pro are proprietary models, so parameter counts are not publicly disclosed. Qwen3.5-397B-A17B is a mixture of experts (MoE) model with 397B total parameters and 17B active parameters.

For model evaluation, we use the following 14 test-taker models:

- openai/gpt-5.4-mini
- nvidia/NVIDIA-Nemotron-Nano-9B-v2
- google/gemma-3-4b-it
- google/gemma-3-12b-it
- Qwen/Qwen3-4B-Instruct-2507
- allenai/Olmo-3-7B-Instruct
- cyankiwi/Olmo-3.1-32B-Instruct-AWQ-4bit
- meta-llama/Llama-3.2-3B-Instruct
- meta-llama/Llama-3.1-8B-Instruct
- LiquidAI/LFM2-2.6B
- LiquidAI/LFM2-1.2B
- LGAI-EXAONE/EXAONE-4.0-1.2B
- ibm-granite/granite-4.0-1b
- ibm-granite/granite-4.0-micro

For evaluation, we specify the following parameters: maximum generation length of 32,768 tokens, no temperature override, two model-call retries, and no stop sequence override. Due to architectural constraints of some models, we modify the max model length for the following: Gemma (124k) and certain LFM & older Qwen3 models (32k). GPT-5.4-mini is run on high reasoning effort and the remaining vLLM-based test-taker models are run on medium reasoning. None of the vLLM models used support reasoning effort, but we enable thinking mode if available.

Test-taker model sizes range from 1B to 12B, with Olmo-3.1-32B-Instruct as the only exception. Additionally, GPT-5.4-mini is proprietary and parameter count is not publicly disclosed.

We use CPUs only when running LLMs via APIs and one NVIDIA RTX A6000 GPU for open-weight LLMs below 8B parameters called via vLLM. We allocate 12 hours for each run, though model jobs are split into separate SLURM jobs by each setting and dataset for parallel speedup. All results are reported from a single run.

Across all generation and evaluation runs, experiments use 657.1M tokens across 377.8 job-hours. Generation runs used a collective 94.9M tokens and 40.5 job-hours. Evaluation used 562.3M tokens across 337.3 job-hours. Token counts include input, reasoning, and output tokens.

We do not perform a hyperparameter search over sampling settings to keep settings consistent between different models in InspectAI. Generation and evaluation settings, as described above, are fixed for final runs.

#### A.3.1 Prompts

After testing prompts, we mainly change the output format to a formal JSON format with a "distractors" list, containing a list of exact choices depending on the input distractor count. For final prompts used in all reported runs, see Prompts A.3 and A.4 for generator model generation prompts and Prompt A.1 for the test-taker model evaluation prompt.

#### A.3.2 Licenses

We follow original licenses and terms of use of all models and other software used. We use InspectAI under its MIT license and vLLM and sentence-transformers/all-MiniLM-L6-v2 under Apache 2.0. Hosted model APIs each have relevant provider terms: OpenAI (GPT-5.2, GPT-5.4-mini), Google (Gemini-3.1-Pro-Preview), and Together AI (Qwen3.5-397B-A17B generation through Together). For open-weight models, we mainly obtain license details from the corresponding HuggingFace model cards. These include licenses such as Apache 2.0 (Qwen, OLMo, IBM Granite), Gemma terms, Llama 3.x terms, and other model-specific licenses for Nemotron-Nano, LFM2, and EXAONE.

### A.4 Position Bias

To analyze the impact of position bias on our results, we show the distribution of test-taker LLM predictions across our answer choices (Tables 4 and 5; $n = 99{,}288$ responses per setting; §4.2). This shows that shuffling the answer choices was a necessary step.

### A.5 Distractor Choices

Figure 6 shows which distractors test-takers choose in each setting (§4.2). Table 6 shows how

| Setting | Metric | A | B | C | D |
|---|---|---|---|---|---|
| Human-guided | Gold answer at position (%) | 23.9 | 25.6 | 26.0 | 24.5 |
| Human-guided | Test-taker predicted (%) | 29.5 | 25.0 | 23.9 | 21.6 |
| LLM-only | Gold answer at position (%) | 24.6 | 26.5 | 24.9 | 24.0 |
| LLM-only | Test-taker predicted (%) | 30.8 | 25.0 | 23.2 | 21.0 |

Table 4: Distribution of test-taker LLM predictions across answer choices in D-GEN ($n = 99{,}288$ responses per setting).

| Setting | Metric | A | B | C | D | E | F | G | H | I | J |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Human-guided | Gold answer at position (%) | 9.9 | 9.8 | 9.6 | 10.7 | 9.8 | 9.6 | 10.1 | 10.3 | 10.7 | 9.4 |
| Human-guided | Test-taker predicted (%) | 19.6 | 12.0 | 12.1 | 10.0 | 8.1 | 7.9 | 8.2 | 7.8 | 6.9 | 7.3 |
| LLM-only | Gold answer at position (%) | 10.6 | 9.9 | 9.7 | 9.6 | 9.5 | 10.8 | 9.3 | 9.9 | 10.6 | 10.0 |
| LLM-only | Test-taker predicted (%) | 20.8 | 12.4 | 12.1 | 9.6 | 8.3 | 7.6 | 7.7 | 7.4 | 7.0 | 7.0 |

Table 5: Distribution of test-taker LLM predictions across answer choices in D-EXT ($n = 99{,}288$ responses per setting).

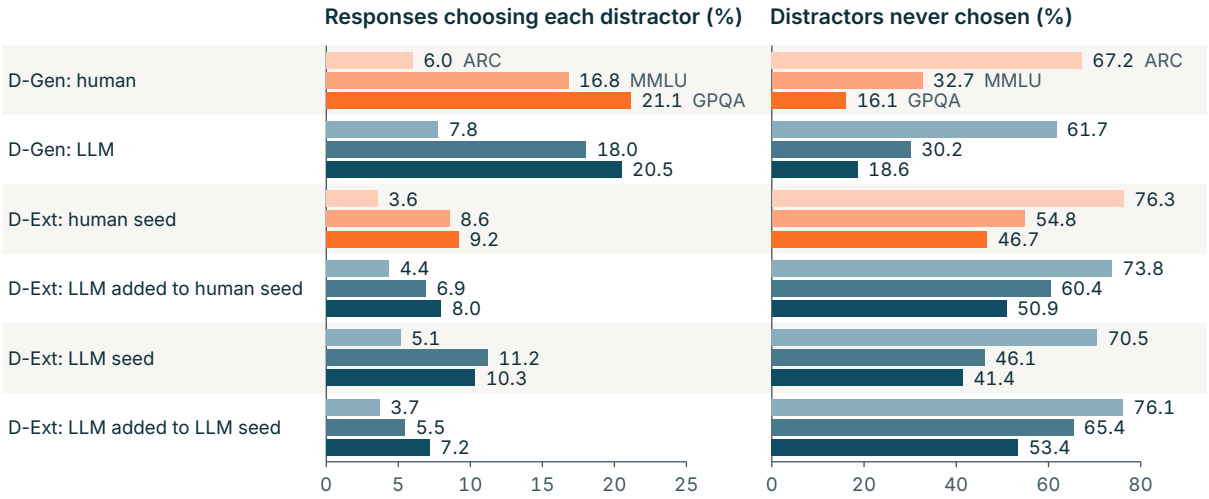



Figure 6: Which distractors test-takers choose, by distractor source and dataset: the share of responses choosing each distractor (left) and the share of distractors no test-taker chooses (right). 74–76% of extension distractors on ARC are never chosen, whether added to human-written or LLM seeds, so never-chosen options can be removed to further compress MCQs for efficiency.

| | | Accuracy (%) | | Chosen distractors | | Top distractor (%) | | Round 2 (%) | |
|---|---|---|---|---|---|---|---|---|---|
| Task | Dataset | HG | LLM | HG | LLM | HG | LLM | HG | LLM |
| D-GEN | ARC | 82.0 | 76.7* | 0.98 | 1.15* | 77 | 76* | – | – |
| | MMLU | 49.6 | 45.9* | 2.02 | 2.10* | 70 | 68 | – | – |
| | GPQA | 36.6 | 38.5 | 2.52 | 2.44* | 61 | 63* | – | – |
| D-EXT | ARC | 63.0 | 62.3 | 2.29 | 2.32 | 59 | 59 | 26.2 | 22.3* |
| | MMLU | 32.9 | 33.5 | 3.73 | 3.69 | 52 | 52 | 41.4 | 32.9* |
| | GPQA | 24.5 | 26.0 | 4.55 | 4.55 | 45 | 45 | 47.8 | 43.0* |

Table 6: How test-takers' choices are distributed over distractors in human-guided (HG) and LLM-only (LLM) items, pooled over the three generators. Accuracy: share of the 14 test-takers' responses on the gold answer. Chosen distractors: distractors per item chosen by at least one test-taker (of 3 in D-GEN, 9 in D-EXT). Top distractor: share of an item's wrong picks on its most-chosen distractor (items with at least three wrong picks). Round 2: share of responses on the six extension distractors. *: the paired difference (same question and generator) has a 95% bootstrap confidence interval over questions that excludes zero.

test-takers' choices are distributed over distractors (§4.2).

### A.6 Choices-Only Accuracy

If distractors leak the answer through surface cues, test-takers can find it without the question (Balepur et al., 2024). We test this with a choices-only prompt (Prompt A.2) on six test-takers (GPT-5.4-mini, Qwen-3 4B, Olmo-3 7B, LLaMA-3 3B and 8B, Nemotron-2): every condition exceeds chance, including 41% on ARC for both human-guided and LLM-only items (Table 7). In D-GEN, LLM-only

| Dataset | Generator | Human-guided | LLM-only | Δ |
|---|---|---|---|---|
| D-GEN *(chance: 25%)* | | | | |
| ARC | GPT | 40.8 | 40.8 | 0.0 |
| | Gemini | 40.8 | 42.2 | +1.4 |
| | Qwen | 40.8 | 42.7 | +1.9 |
| MMLU | GPT | 33.8 | 27.1 | **−6.7** |
| | Gemini | 33.8 | 31.8 | −2.0 |
| | Qwen | 33.8 | 36.4 | +2.5 |
| GPQA | GPT | 30.0 | 32.8 | +2.9 |
| | Gemini | 30.0 | 32.9 | +3.0 |
| | Qwen | 30.0 | 38.0 | **+8.0** |
| D-EXT *(chance: 10%)* | | | | |
| ARC | GPT | 27.0 | 29.5 | **+2.6** |
| | Gemini | 29.5 | 32.1 | **+2.6** |
| | Qwen | 31.7 | 33.4 | +1.7 |
| MMLU | GPT | 18.1 | 18.1 | 0.0 |
| | Gemini | 23.0 | 18.6 | **−4.4** |
| | Qwen | 23.9 | 22.1 | −1.7 |
| GPQA | GPT | 20.0 | 22.7 | +2.7 |
| | Gemini | 18.0 | 16.9 | −1.1 |
| | Qwen | 22.4 | 26.2 | +3.9 |

Table 7: Every condition exceeds chance, including 41% on ARC for both human-guided and LLM-only items. Choices-only accuracy (%) of six test-takers; bold differences are significant after a Holm correction.

items differ from human-guided ones in only 2 of 9 pairs (Qwen on GPQA, +8.0 points; GPT on MMLU, −6.7), and in D-EXT, in 3 of 9 (GPT and Gemini on ARC, +2.6 each; Gemini on MMLU, −4.4). Still, LLM-only items score higher in 11 of 18 pairs, so LLM distractors slightly add shortcuts. These are the same two D-GEN sets that are significantly more discriminative (GPT on MMLU) and less (Qwen on GPQA; Figure 7), consistent with leaked answers helping weaker test-takers. Option length is another surface cue: the gold answer is the longest choice in 26–27% of human-guided D-GEN items but only about 10% of LLM-only ones, yet test-takers pick the longest choice less often on LLM-only items (20–25% vs. 31–35% of responses on ARC and MMLU), so neither leaked answers nor length cues explain the D-GEN difficulty gains.

### A.7 IRT Details

The 1.702 scaling factor in §3.4 is a standard constant to minimize the maximum difference between the normal and logistic distribution functions (Camilli, 1994). We fit the model using penalized maximum likelihood with L-BFGS-B, with Gaussian priors on stem effects (SD of 3), item residuals (SD of 1), log-discrimination (SD of 0.75), and logit-guessing (SD of 1 centered at $\text{logit}(1/n)$). Raw accuracy agrees with IRT difficulty ($r = -0.90$ across comparisons). Table 11 shows which D-GEN difficulty differences stay significant across our 18 IRT refits (§3.4).

| Rank | Test-taker model | Relative ability |
|---|---|---|
| 1 | GPT-5.4-mini | 2.591 |
| 2 | Nemotron-Nano-9B-v2 | 1.349 |
| 3 | Gemma-3-12B-it | 0.543 |
| 4 | Qwen3-4B-Instruct | 0.487 |
| 5 | Olmo-3.1-32B-Instruct-AWQ | 0.361 |
| 6 | Llama-3.1-8B-Instruct | −0.002 |
| 7 | LFM2-2.6B | −0.219 |
| 8 | EXAONE-4.0-1.2B | −0.358 |
| 9 | Gemma-3-4B-it | −0.447 |
| 10 | Olmo-3-7B-Instruct | −0.487 |
| 11 | Granite-4.0-micro | −0.557 |
| 12 | Llama-3.2-3B-Instruct | −0.622 |
| 13 | Granite-4.0-1B | −1.050 |
| 14 | LFM2-1.2B | −1.589 |

Table 8: Relative abilities of each test-taker model based on $\theta_j$ parameter. Ability estimates are centered by subtracting the mean ability across the 14 test-taker models.

| Dataset | Source stems | Difficulty effect |
|---|---|---|
| GPQA | 415 | −0.046 |
| MMLU | 954 | −0.723 |
| ARC-Challenge | 995 | −2.493 |

Table 9: Relative difficulties of each dataset based on $\beta^{\text{dataset}}_{d(i)}$ sub-parameter. Larger estimates correspond to harder datasets.

| Setting | Difficulty effect |
|---|---|
| Human-guided D-EXT | 0.611 |
| LLM-only D-EXT | 0.597 |
| All-at-once D-EXT | 0.430 |
| LLM-only D-GEN | 0.139 |
| Human-guided D-GEN | 0.000 |

Table 10: The two-step methods outperform the one-step ablation in overall MCQ difficulty. Setting-level IRT difficulty effects based on $\beta^{\text{setting}}_{r(i)}$ sub-parameter. Using human-guided D-GEN as the reference setting, larger estimates indicate that a setting increases item difficulty relative to the reference.

We also include fit details:

- Observations: 496,440
- Test-taker models: 14
- Unique stems: 2,364
- Instantiated items: 35,460
- Parameters: 108,764
- Optimizer: L-BFGS-B
- Converged: YES, after 1,870 iterations
- Log likelihood: -169915.14
- AIC: 557358.27
- BIC: 1766293.83

Additionally, we provide per-parameter results: Tables 8 (test-taker), 9 (dataset), and 10 (setting).

### A.8 Full Quality Results

We report the full details of the delta differences shown in Figure 3 in Figure 8. Difficulty and dis-

| Fit | ARC | | | MMLU | | | GPQA | | |
|---|---|---|---|---|---|---|---|---|---|
| | GPT | Gemini | Qwen | GPT | Gemini | Qwen | GPT | Gemini | Qwen |
| Main fit | + | + | + | + | + | ○ | + | + | ○ |
| Leave one test-taker out (13 of 14 refits) | + | + | + | + | + | ○ | + | + | ○ |
| Leave out Nemotron-2 | + | + | + | + | + | − | ○ | ○ | ○ |
| No random-fallback answers | + | + | + | + | + | ○ | + | + | ○ |
| No web-flagged ARC stems | + | + | + | + | + | ○ | + | + | ○ |
| Human copies share one generator effect | + | + | + | + | + | ○ | + | ○ | ○ |
| Generator effects fit per dataset | + | + | + | + | + | ○ | ○ | + | − |

Table 11: Significance of each D-GEN difficulty difference (+ harder, ○ not significant, − easier; Holm-corrected) in the main fit and 18 refits.

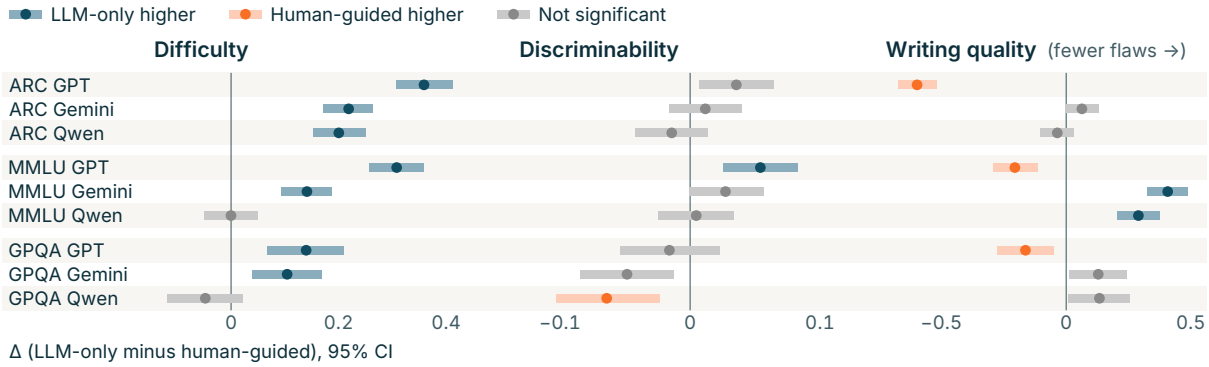



Figure 7: Differences in difficulty, discriminability, and writing flaws between LLM-only and human-guided items in D-GEN with 95% confidence intervals (for writing quality, positive differences mean fewer flaws); colors show which differences stay significant after a Holm correction. 7 of 9 LLM distractor sets are significantly harder than human-guided items, but only one is significantly more discriminative (GPT on MMLU): harder LLM distractors are not always better.

| Statistic | Value |
|---|---|
| Mean flaws/item | $2.641 \pm 0.009$ |
| SD | 1.690 |
| Median | 2 |
| IQR | 1–4 |
| Range | 0–12 |
| Zero flaws | 7.7% |
| $\geq 2$ flaws | 73.0% |

Table 12: Overall flaw-count distribution.

criminability are estimated with the IRT model described in Appendix A.7; writing quality is the average number of BenchMarker writing flaws per item.

## A.9 Full Writing-Flaw Results

Balepur et al. (2026) validate BenchMarker's judgments against human annotations: two human annotators rated 50 random items for each of the 19 writing flaws and reached 85% agreement for every writing flaw. LLMs also achieve high agreement: 0.82 accuracy ($\kappa$=.53) for Gemini 2.5 Pro and 0.81 ($\kappa$=.50) for GPT-5. Tables 12 and 13 have general flaw-count information and a full BenchMarker writing-flaw breakdown. The LLMs used to evaluate each of the 19 writing flaws are in Table 15 (Balepur et al., 2026). We break down the five writing flaws by generator model in Table 16.

## A.10 Do Writing Flaws Explain Difficulty?

We regress item IRT difficulty and discriminability on the 19 writing flaws over all 35,460 items (Table 17). Effects vary by flaw type—multiple best answers make items harder and less discriminative, while implausible distractors and negatives make them easier. Controlling for flaws, GPT's D-GEN difficulty gain decreases by 13–20% (ARC: 0.36 → 0.29; MMLU: 0.31 → 0.27; GPQA: 0.14 → 0.11). This is mostly due to multiple best answers, though all three gains remain significant.

## A.11 Qualitative Analysis Details

All four annotators are NLP researchers. Table 14 has per-strategy agreement among the four annotators. We define the five annotated strategies as:

**SOLVE** The model re-derives the solution for the gold answer from the question step by step.

**SIMULATE** The model attempts to simulate test-taker mistakes such as wrong calculation, formula misuse, etc. Then, the model assigns each distractor to a specific error.

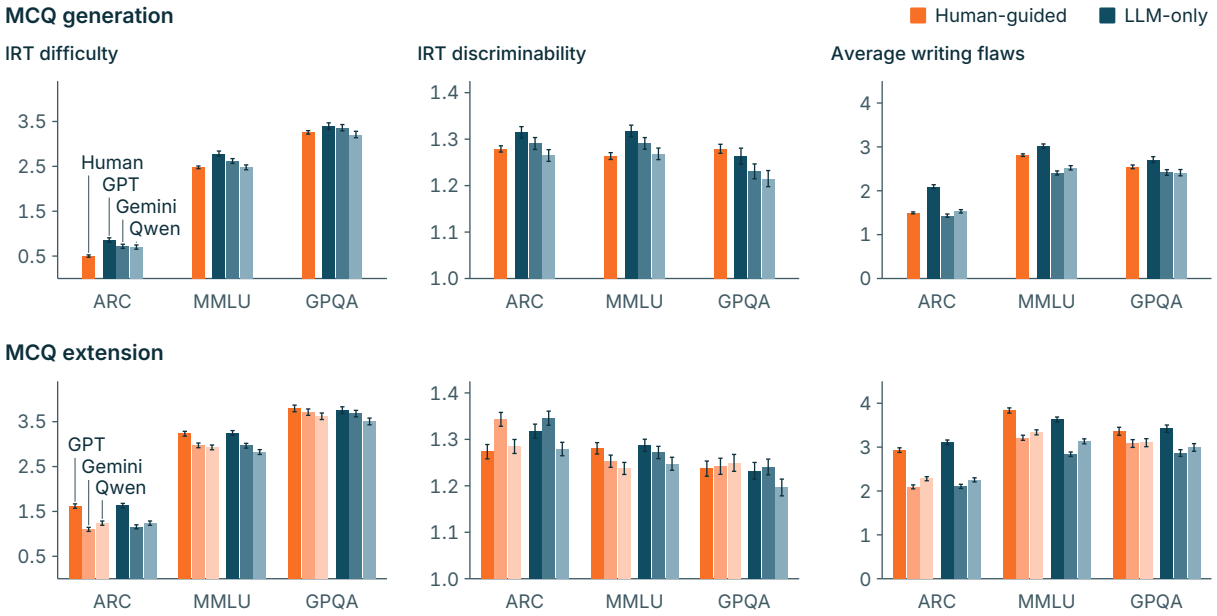



Figure 8: Per-model breakdown of MCQ question quality across three metrics: IRT difficulty (how hard is the question), IRT discriminability (how well does the question tell test-takers apart), and average writing flaws (a proxy for question validity). On every dataset and in both tasks, GPT-based distractors are the most difficult and have the most writing flaws: GPT writes harder distractors at the expense of writing quality.

| | D-GEN | | D-EXT | |
|---|---|---|---|---|
| Writing Flaw Type | Human | Model | Human | Model |
| Unequal Option Length | $\mathbf{20.4 \pm 0.48}$ | $28.0 \pm 0.53$ | $37.5 \pm 0.57$ | $\mathbf{37.0 \pm 0.57}$ |
| Implausible Distractors | $11.2 \pm 0.37$ | $\mathbf{7.8 \pm 0.32}$ | $22.4 \pm 0.50$ | $\mathbf{19.3 \pm 0.47}$ |
| Has Absolute Terms | $\mathbf{9.6 \pm 0.35}$ | $14.4 \pm 0.42$ | $\mathbf{23.2 \pm 0.50}$ | $23.7 \pm 0.50$ |
| Unclear Language | $8.7 \pm 0.33$ | $\mathbf{6.9 \pm 0.30}$ | $11.2 \pm 0.37$ | $\mathbf{9.2 \pm 0.34}$ |
| Multiple Best Answers | $\mathbf{12.2 \pm 0.39}$ | $15.3 \pm 0.43$ | $29.3 \pm 0.54$ | $\mathbf{28.6 \pm 0.54}$ |
| Has Convergence Cues | $21.6 \pm 0.49$ | $\mathbf{19.1 \pm 0.47}$ | $20.6 \pm 0.48$ | $\mathbf{19.3 \pm 0.47}$ |
| Has Extraneous Information | $52.7 \pm 0.59$ | $\mathbf{51.4 \pm 0.59}$ | $\mathbf{50.0 \pm 0.59}$ | $50.4 \pm 0.59$ |
| Unfocused Stem | $\mathbf{28.4 \pm 0.54}$ | $28.5 \pm 0.54$ | $28.6 \pm 0.54$ | $28.6 \pm 0.54$ |
| Problem Not in Stem | $12.2 \pm 0.39$ | $\mathbf{11.4 \pm 0.38}$ | $11.2 \pm 0.37$ | $11.2 \pm 0.38$ |
| Unordered Options | $11.3 \pm 0.38$ | $\mathbf{11.2 \pm 0.37}$ | $\mathbf{11.3 \pm 0.38}$ | $11.5 \pm 0.38$ |
| Has Vague Terms | $5.6 \pm 0.27$ | $5.6 \pm 0.27$ | $\mathbf{12.2 \pm 0.39}$ | $13.0 \pm 0.40$ |
| Has Logical Cues | $6.1 \pm 0.28$ | $\mathbf{4.4 \pm 0.24}$ | $11.7 \pm 0.38$ | $\mathbf{10.0 \pm 0.36}$ |
| Grammatical Inconsistency | $5.1 \pm 0.26$ | $\mathbf{4.4 \pm 0.24}$ | $8.6 \pm 0.33$ | $\mathbf{7.3 \pm 0.31}$ |
| Has K-Type Options | $3.0 \pm 0.20$ | $\mathbf{2.4 \pm 0.18}$ | $9.5 \pm 0.35$ | $\mathbf{8.1 \pm 0.32}$ |
| Has Repetition | $5.9 \pm 0.28$ | $\mathbf{4.0 \pm 0.23}$ | $\mathbf{2.9 \pm 0.20}$ | $3.4 \pm 0.21$ |
| Has Negatives | $3.8 \pm 0.23$ | $3.8 \pm 0.23$ | $\mathbf{3.5 \pm 0.22}$ | $3.6 \pm 0.22$ |
| Has Fill-in-the-Blank | $3.2 \pm 0.21$ | $\mathbf{3.1 \pm 0.21}$ | $3.3 \pm 0.21$ | $3.3 \pm 0.21$ |
| Has None of the Above | $\mathbf{0.1 \pm 0.04}$ | $0.2 \pm 0.05$ | $0.7 \pm 0.10$ | $\mathbf{0.6 \pm 0.09}$ |
| Has All of the Above | $0.1 \pm 0.04$ | $0.1 \pm 0.03$ | $0.3 \pm 0.06$ | $\mathbf{0.2 \pm 0.05}$ |

Table 13: Prevalence (%) of all 19 writing flaws in human-guided and LLM-only distractors across distractor generation and extension tasks. Values are BenchMarker rule failure rates, so lower is better.

| Strategy | $\kappa$ [95% CI] | All agree | AC1 |
|---|---|---|---|
| SOLVE | 0.70 [0.56, 0.83] | 80% | 0.83 |
| SIMULATE | 0.60 [0.45, 0.74] | 70% | 0.69 |
| RECALL | 0.68 [0.55, 0.81] | 72% | 0.68 |
| SHALLOW | 0.52 [0.37, 0.66] | 57% | 0.54 |
| REFINE | 0.54 [0.36, 0.69] | 68% | 0.73 |
| All decisions | 0.66 [0.60, 0.72] | 69% | 0.67 |

Table 14: Agreement (Fleiss' $\kappa$) among four annotators labeling reasoning-trace strategies on 60 GPQA D-GEN traces (20 per generator); 95% confidence intervals from a bootstrap over traces.

**RECALL** The model creates distractors based on related-but-distinct subject information, such as alternative reaction products, adjacent elements/compounds, parallel mechanisms, etc.

**SHALLOW** The model makes simple perturbations of the correct answer, such as swapping signs, adjacent integers, rearranging, etc.

**REFINE** The model reflects over the entire set and critiques distractors for aspects like distinctness, plausibility, and format. Then, it revises the distractors.

We also provide the full trace of the Gemini ques-

| Judge LLM | Writing flaws |
|---|---|
| google/gemini-2.5-pro | avoid_repetition, equal_length_options, focused_stem, grammatical_consistency, no_absolute_terms, no_fill_in_blank, ordered_options |
| google/gemini-2.5-flash | no_none_of_the_above, no_vague_terms, single_best_answer |
| google/gemini-2.5-flash-lite | avoid_k_type, problem_in_stem |
| anthropic/claude-sonnet-4-5-20250929 | avoid_negatives, no_all_of_the_above, no_convergence_cues |
| openai/gpt-5-2025-08-07 | clear_language, no_extraneous_info, no_logical_cues, plausible_distractors |

Table 15: BenchMarker judge LLM for each of the 19 writing flaws (Balepur et al., 2026).

| | D-GEN | | | D-EXT | | |
|---|---|---|---|---|---|---|
| Writing Flaw Type | Gemini | GPT | Qwen | Gemini | GPT | Qwen |
| Unequal Option Length | $-4.3^{*}$ | $+27.7^{*}$ | $-0.6$ | $-3.3^{*}$ | $+5.2^{*}$ | $-3.5^{*}$ |
| Implausible Distractors | $-5.8^{*}$ | $-1.4$ | $-2.9^{*}$ | $-3.6^{*}$ | $-2.5^{*}$ | $-3.4^{*}$ |
| Has Absolute Terms | $+2.2^{*}$ | $+6.5^{*}$ | $+5.7^{*}$ | $+0.4$ | $+0.2$ | $+0.9$ |
| Multiple Best Answers | $-0.3$ | $+7.7^{*}$ | $+1.8^{*}$ | $-1.9^{*}$ | $+0.8$ | $-1.1$ |
| Has Convergence Cues | $-1.8$ | $-5.5^{*}$ | $-0.3$ | $-1.8$ | $-1.8$ | $-0.2$ |

Table 16: Percentage of five writing flaws in LLM-only minus human-guided distractors in D-GEN and D-EXT per generator model (Table 2), in percentage points. Positive differences mean more writing flaws in LLM distractors. $^{*}$: the 95% confidence interval from a paired bootstrap over question stems excludes 0 (unadjusted).

| Writing Flaw Type | Difficulty | Discriminability |
|---|---|---|
| Flaw count (per flaw) | $\mathbf{0.047 \pm 0.004}$ | $-0.003 \pm 0.002$ |
| Unequal Option Length | $\mathbf{0.048 \pm 0.009}$ | $0.016 \pm 0.006$ |
| Implausible Distractors | $\mathbf{-0.036 \pm 0.012}$ | $0.022 \pm 0.007$ |
| Has Absolute Terms | $\mathbf{-0.056 \pm 0.012}$ | $0.011 \pm 0.008$ |
| Unclear Language | $\mathbf{0.074 \pm 0.018}$ | $-0.008 \pm 0.011$ |
| Multiple Best Answers | $\mathbf{0.403 \pm 0.015}$ | $\mathbf{-0.089 \pm 0.007}$ |
| Has Convergence Cues | $0.010 \pm 0.012$ | $-0.011 \pm 0.007$ |
| Has Extraneous Information | $-0.022 \pm 0.011$ | $0.002 \pm 0.007$ |
| Unfocused Stem | $0.048 \pm 0.045$ | $-0.007 \pm 0.024$ |
| Problem Not in Stem | $\mathbf{0.079 \pm 0.018}$ | $0.006 \pm 0.011$ |
| Unordered Options | $0.044 \pm 0.030$ | $0.041 \pm 0.019$ |
| Has Vague Terms | $0.045 \pm 0.016$ | $0.018 \pm 0.009$ |
| Has Logical Cues | $-0.021 \pm 0.016$ | $-0.011 \pm 0.010$ |
| Grammatical Inconsistency | $0.036 \pm 0.017$ | $0.024 \pm 0.011$ |
| Has K-Type Options | $0.011 \pm 0.017$ | $-0.014 \pm 0.011$ |
| Has Repetition | $-0.049 \pm 0.024$ | $-0.001 \pm 0.015$ |
| Has Negatives | $\mathbf{-0.175 \pm 0.040}$ | $-0.023 \pm 0.031$ |
| Has Fill-in-the-Blank | $0.054 \pm 0.058$ | $0.064 \pm 0.044$ |
| Has None of the Above | $-0.090 \pm 0.047$ | $-0.006 \pm 0.035$ |
| Has All of the Above | $-0.004 \pm 0.083$ | $-0.091 \pm 0.057$ |

Table 17: Change in item IRT difficulty and discriminability per writing flaw. Estimates use stem fixed effects and $\pm$ SEs clustered by stem, and bold values are significant after Holm correction. Multiple best answers make items harder and less discriminative, while other flaws can make items easier.

tion referred to in §5.3: Trace A.1.

### A.12 Two-Step Extension Makes Items Harder, Not More Discriminative

In the LLM-only D-EXT task, the generator model adds six distractors to an MCQ with three distractors previously generated in D-GEN. This spurs us to ask whether there are benefits to generating all nine distractors at once. The two-step process leads to harder MCQs without sacrificing discriminability or writing quality (Figure 9), but they are not more discriminative either, so, as in §4.1, harder items are not always better.




Figure 9: Per-model measurements of difficulty, discriminability, and writing quality for the extension setting. Two-step items are harder than one-step items, with similar discriminability and writing quality.

Question QA.1: MMLU-Pro:996 (Text Corruption)

**Question:** A homeowner resides downhill from a metal fabrication facility. She has sued both the owner of the facility and the supplier of a solvent used at the facility. She contends that contaminants, consisting mostly of the solvent, were released into the ground at the facility and have migrated and continue to migrate to her property, contaminating the soil, the groundwater, and her well. She alleges various acts of negligence on the part of the facility owner in causing the release of the contaminants into the ground. She also alleges that employees of the solvent supplier were negligent in frequently the solvent supplier were negligent in frequently spilling some of the solvent onto the ground while filling a rooftop tank at the facility. The solvent supplier has moved for summary judgment, arguing that if there was any contamination, the facility owner and the supplier independently contributed indeterminate amounts to the contamination and that therefore the homeowner cannot show how much damage each has inflicted on her. There is no evidence that the facility owner and the solvent supplier acted in concert. Should the court grant the summary judgment motion?

**Choices:**
A. No, because concurrent tortfeasors are jointly and severally liable for an indivisible injury.
B. No, because the solvent supplier is vicariously liable for damage inflicted by the facility owner.
C. Yes, because there is no basis for allocating damages against the solvent supplier. damages against the solvent supplier.
D. Yes, because there is no evidence that the facility owner and the solvent supplier acted in concert.

**Correct Answer:** A

Question QA.2: GPQA:recDjE01bu72pPUU2 (Duplicate correct answer among distractors)

**Question:**
When 500 mL of PH3 is decomposed the total volume of the reaction mixture becomes 600 mL only. The H2 obtained in the above reaction is used to create electricity in a fuel cell. Calculate the volume of unreacted H2 in the fuel cell when only 50 mL of O2 is used.
**Choices:**
A. 200 mL
B. 100 mL
C. 180 mL
D. 200 mL
**Correct Answer:** A

Question QA.3: GPQA:recZSGUkn56v9kEp1 (Duplicate distractor option)

**Question:**
Determine the product obtained when 3,4-dimethylhexanedial is reacted with the following set of reagents:
1. KOH, H2O, THF, Heat
2. CH3CH2MgBr, H3O+
3. PCC, CH2Cl2
4. O3, H2O
**Choices:**
A. 4,5-dimethylnonane-2,6,7-trione
B. 4,5-dimethylnonane-2,6,7-trione
C. 3,4-dimethyl-5,6-dioxooctanal
D. 3,4-dimethyl-5,6-dioxooctanoic acid
**Correct Answer:** D

Question QA.4: GPQA:recnGEpF1srQpaqWq (Duplicate correct answer among distractors)

**Question:**
Which of the following molecules is formed when 5-butylnona-2,6-diene is heated?
**Choices:**
A. 5-ethylundeca-2,6-diene
B. 5-ethyl-4-methyldeca-2,6-diene
C. 4-ethyl-3-methyldeca-1,5-diene
D. 5-ethyl-4-methyldeca-2,6-diene
**Correct Answer:** C

Question QA.5: GPQA:recnTTKdBzfuoZ7w7 (Impractically long DNA sequence)

**Question:**
Congratulations, you just landed your dream job at one of the top structural biology labs at Stanford University in California. Without further ado, the Principal Investigator explains that the priority of the moment is to produce in E. coli BL21 then purify as quickly as possible the Human P53 protein.
The amino acid sequence of this protein is as follows:
MEEPQSDPSVEPPLSQETFSDLWKLLPENNVLSPLPSQAMDDLMLSPDDIEQWFTEDPGPDEA
PRMPEAAPPVAPAPAAPTPAAPAPAPSWPLSSSVPSQKTYQGSYGFRLGFLHSGTAKSVTCTY
SPALNKMFCQLAKTCPVQLWVDSTPPPGTRVRAMAIYKQSQHMTEVVRRCPHHERCSDSDGLA
PPQHLIRVEGNLRVEYLDDRNTFRHSVVVPYEPPEVGSDCTTIHYNYMCNSSCMGGMNRRPIL
TIITLEDSSGNLLGRNSFEVRVCACPGRDRRTEEENLRKKGEPHHELPPGSTKRALPNNTSSS
PQPKKKPLDGEYFTLQIRGRERFEMFRELNEALELKDAQAGKEPGGSRAHSSHLKSKKGQSTS
RHKKLMFKTEGPDSD

You are given 4 plasmids containing 4 different sequences, which one are you going to use?
**Choices:** (cut off for brevity)
A. ATGGAGGAGCCGCAGTCAGATCCTAGCGTCGAGCCGCCTCTGAGTCAGGA…
B. ATGGAGGAGCCGCAGTCAGATCCTAGCGTCGAGCCGCCTCTGAGTCAGGA…
C. ATGGAGGAGCCGCAGTCAGATCCTAGCGTCGAGCCCCCTCTGAGTCAGGA…
D. ATGGAGGAGCCGCAGTCAGATCCTAGCGTCGAGCCGCCTCTGAGTCAGGA…
**Correct Answer:** C

Prompt A.1: Evaluation

Answer the following multiple choice question.

<question>
question
</question>

<choices>
choices
</choices>

<format>
Return your output as valid JSON with the key “answer” which is one of letters.
json_example
Do not include anything else.
</format>

**Prompt A.2: Evaluation (Choices-Only)**

Answer the following multiple choice question just by using the choices and without access to the question. Use any strategy possible to come up with the correct answer.

<choices>
choices
</choices>

<format>
Return your output as valid JSON with the key "answer" which is one of letters.
json_example
Do not include anything else.
</format>

**Prompt A.3: Generate given QA pair (Generation Setting)**

You are an expert writer of plausible but incorrect distractors for multiple-choice questions (MCQs).

Given a question stem and an answer to this question, your job is to generate exactly count plausible but incorrect answer choices that could be used as distractors to form a full multiple-choice question.

Here is the question stem and the correct answer:
<multiple-choice question>
Question: question
Answer: gold_answer
</multiple-choice question>

<distractor instructions>
- Each distractor option must be a plausible answer to this question that tests common misconceptions, but ultimately should not be more correct than the provided answer
- Each of the distractor options must be distinct; they cannot be synonyms of each other
</distractor instructions>

<format>
Return your output as valid JSON with exactly one key: "distractors" - a list of exactly count incorrect answer choices
json_example
Do not include anything else.
</format>

**Prompt A.4: Generate given MCQA (Extension Setting)**

You are an expert writer of plausible but incorrect distractors for multiple-choice questions (MCQs).

Given a multiple-choice question stem, old_count existing choices for this question, and an answer to this question, your job is to generate exactly count additional plausible but incorrect answer choices to form an expanded multiple-choice question.

Here is the question stem, the choices, and the correct answer:
<multiple-choice question>
Question: question
Choices:
choices
Answer: gold_answer
</multiple-choice question>

<distractor instructions>
- Each distractor option must be a plausible answer to this question that tests common misconceptions, but ultimately should not be more correct than the provided answer
- Each of the distractor options must be distinct; they cannot be synonyms of each other
- None of the generated distractor choices should be synonyms with the provided answer choices above.
</distractor instructions>

<format>
Return your output as valid JSON with exactly one key: "distractors" - a list of exactly count incorrect answer choices
json_example
Do not include anything else.
</format>

Trace A.1: §5.3 Question (gpqa:rec2xN0SZSiyfKb4E)

**Analyzing AGN Torus Radius Calculations and Potential Student Errors**
Okay, so we're looking at calculating the inner radius of an AGN torus using the Barvainis model, given a dust sublimation temperature of 900 K, a bolometric luminosity of $1.5 \times 10^{46}$ erg/s, and a UV bolometric correction of 5. The correct answer, we're told, is 3 pc. Let's break this down systematically.
The Barvainis model is the key here; it relates the dust sublimation radius to the UV luminosity. The formula is $R \approx 1.3 \left(\frac{L_{UV}}{10^{46}\text{ erg/s}}\right)^{0.5} \left(\frac{T}{1500\text{ K}}\right)^{-2.8}$ pc. First, we need to calculate the UV luminosity from the bolometric luminosity: $L_{UV} = L_{\text{bol}}/BC = (1.5 \times 10^{46})/5 = 0.3 \times 10^{46} = 3 \times 10^{45}$ erg/s. Let's make sure we are clear about the parameters. Then we plug that into the formula.
Substituting the values into the Barvainis formula, we have $L_{UV,46} = 0.3$, $T = 900$ K, so $T_{1500} = 900/1500 = 0.6$. The radius becomes $R \approx 1.3 \times \sqrt{0.3} \times (0.6)^{-2.8}$. Let's calculate this step by step. $\sqrt{0.3} \approx 0.5477$, and $0.6^{-2.8} \approx 4.18$. Multiplying everything together, we get $R \approx 1.3 \times 0.5477 \times 4.18 \approx 2.97$ pc. That is almost exactly 3 pc.
Now, let's play "what if" and consider what common errors a student might make. If they forget to divide by the bolometric correction (a likely scenario), they'd use the bolometric luminosity directly. In that case, $L = 1.5 \times 10^{46}$ erg/s, and $R = 1.3 \times \sqrt{1.5} \times (0.6)^{-2.8} \approx 6.66$, or around 6.7 pc.
If the student uses the bolometric correction backwards and multiplies, we get $L = 1.5 \times 10^{46} \times 5 = 7.5 \times 10^{46}$. The radius would become $R = 1.3 \times \sqrt{7.5} \times (0.6)^{-2.8} \approx 14.89$ pc, or about 15 pc.
If they forget the temperature dependence entirely, or assume a standard 1500K temperature, the calculation simplifies to $R = 1.3 \times \sqrt{0.3} \approx 0.71$, which is about 0.7 pc.
We can consider the Stefan-Boltzmann law and its $T^{-2}$ relation, instead of the Barvainis $T^{-2.8}$. If this happens, we get $R = 1.3 \times \sqrt{0.3} \times (0.6)^{-2} = 1.3 \times \sqrt{0.3} \times 2.77$. Thus $R \approx 1.97$, which is about 2 pc.
Finally, what are plausible wrong answers (distractors)? We have 6.7 pc (no bolometric correction), 15 pc (multiplying by the BC), 0.7 pc (no temperature dependence), and we've already done 1.6 pc (forgetting the square root) and 2 pc. The distractors should also be formatted the same. I will go with "6.7 pc", "15 pc", and "0.7 pc".